QCert: Verifying and Extending Quantum Circuit Completeness in Agda and Lean

ANONYMOUS AUTHOR(S)

A set of circuit equations is complete if every equation between circuits denoting the same linear map follows from it, and minimal if no equation can be derived from the others. Establishing these properties takes long case analyses spanning hundreds of pages, making the details hard to check by hand. To address this, we extend an existing formalization library to QCert in two proof assistants. In QCert, humans set up a unified framework of circuits, matrix semantics, and group presentations, and state each paper's main theorem using these definitions. LLM coding agents then write the proofs following the papers' arguments, and the proof assistants check them. A human therefore need only review each paper's relations, gate semantics, and theorem statement. Using QCert, we first check sixteen published results, fifteen on completeness and one on minimality. They cover discrete gate sets such as Clifford, Clifford+T, and Toffoli–Hadamard, on qubits or qudits of every odd prime dimension. We verify fourteen of them directly and correct an error in each of the other two. We also obtain a complete equational theory for circuits over $\{X, CX, CCX\}$, and extend the completeness of Toffoli–Hadamard circuits from three qubits to any number of qubits. We expect QCert to help with other open completeness problems, such as those for quantum circuits in higher dimensions.

1 Introduction

Rewriting of quantum circuits is a ubiquitous tool in circuit optimization [41, 50, 51, 65], equivalence checking [66], hardware mapping [63], and reasoning about quantum programs [31]. This relies on an *equational theory* for a family of quantum circuits, a set of rewrite rules that transform a circuit into an equivalent one. Such a theory should be *sound* so that every rewrite preserves the underlying linear map. It should also be *complete* so that any equivalence between circuits can be derived. Additionally, we want it to be *minimal*, with no redundant rules. Completeness and minimality proofs often involve long case analyses, with supporting derivations spanning hundreds of pages [24, 31, 44]. When a derivation takes many rewrite steps, one step can easily rely, through earlier lemmas, on the very equation being derived. A *proof assistant* can check these dependencies and reject such circular arguments.

Our Contributions. We present QCert, a framework in Agda [19, 53] and Lean [29] for representing and verifying quantum circuit completeness and minimality results. Figure 1 summarises its workflow: humans check if the formal definitions, relations, and theorem statement faithfully express the source paper; LLM coding agents formalize the proof, and Lean or Agda checks it.

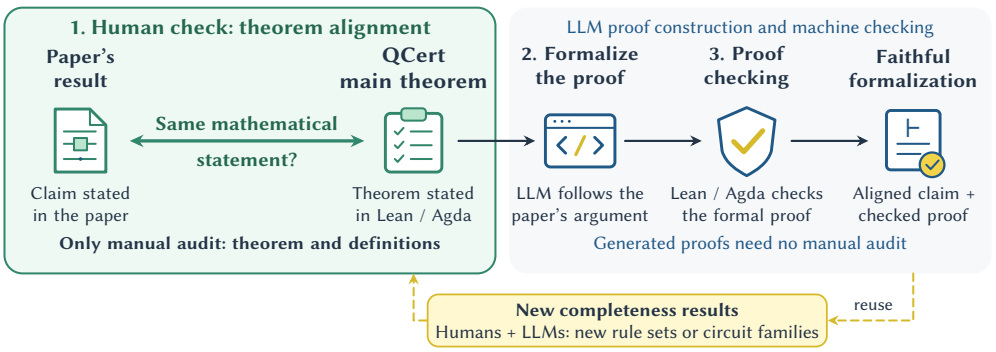


Fig. 1. QCert's workflow. The main path checks published results; the dashed branch reuses the verified libraries for human-guided, LLM-assisted work on new completeness results. Proposed results undergo the same human review of definitions and theorem statements, and the same machine checking of formal proofs.

The manual audit matters because a proof assistant checks a proof only against the theorem as stated, which may not match the intended result [40]. For example, Bastounis et al. [8] find that, because of mistranslations, a claimed Navier–Stokes formalization verified a weaker statement than intended. QCert therefore states each paper's main theorem in its own module, in both Agda and Lean. Since the proof assistant checks every proof, a reviewer need only compare these statements, and the definitions they use, with the source papers to confirm that a formalization is faithful.

QCert builds on two earlier Agda developments. The first is the formalization [11] of the multi-qudit Clifford completeness result of Bian, Li et al. [12]. The second is the library of circuits and group presentations of Bian and Feng [10], which QCert generalizes by adding matrix semantics, minimality, three-wire gates, and scalar generators. Once the Agda framework was complete, we instructed LLM coding agents to translate it into Lean, which has a large community that uses it to formalize mathematics [54, 64], build AI theorem provers [67], and formalize quantum information theory and quantum programs [9, 42, 62, 70]. As a result, QCert can build on Lean's well-developed mathematical library Mathlib. In the long run, the Lean community can use QCert's verified rule sets and completeness theorems to justify circuit rewrites in a verified quantum program.

We then use QCert to check sixteen papers, which include most of the completeness results for quantum circuits over discrete gate sets. We verify fourteen of them and correct errors that peer review had missed in the other two. Most of these circuit families are restricted Clifford+T circuits. Amy et al. [2] show that several of these circuit families, with at most one ancilla, represent exactly the unitary matrices over a subring of $\mathbb{Z}[1/\sqrt{2}, i]$. Figure 2 displays these gate sets, their matrix groups, and multi-level generators. Because these families are nested and their matrices all have entries in $\mathbb{Z}[1/\sqrt{2}, i]$, QCert reuses exact arithmetic, matrix semantics, group constructions, and completeness arguments across formalizations.

Beyond verifying published results, QCert facilitates the discovery of new ones by freeing researchers from labour-intensive checking so they can focus on developing new techniques. Until now, Toffoli–Hadamard circuits had a complete equational theory only on three qubits [3], and its rule set omits a relation that QCert shows to be independent. Working with LLM coding agents, we combine the presentation of $O_n(\mathbb{Z}[1/2])$ by Li et al. [45] with the method of Clément [24] to extend this completeness to any number of qubits. Along the way, we obtain a complete equational theory for circuits over $\{X, CX, CCX\}$. QCert then checks and confirms both proofs, building on its existing formalizations of the presentation and the method they combine.

Background and Related Work. Completeness results have been established for a wide range of circuit families, including reversible linear circuits [43, 58], reversible circuits over multiple-control Toffoli gates [32, 33], Clifford [12, 44, 49, 60], CNOT-dihedral [1, 17], two-qubit Clifford+T [14], three-qubit Clifford+CS [15], Toffoli–Hadamard [3], and real Clifford+CH [24] circuits, as well as universal quantum circuits [26, 27]. Completeness has also been obtained for the group of two-level matrices underlying several of these circuit families [13, 23, 31, 36, 45]. Minimal theories have been given for six of these circuit fragments [18] and for all quantum circuits [25]. Their proofs typically follow a simple strategy, such as rewriting every circuit into a unique normal form, but carrying it out requires overwhelming case distinctions [24, 31, 44].

Peer review cannot check proofs of this volume in detail, so it is desirable to verify them in a proof assistant such as Lean or Agda. This would not have been feasible in the past due to the sheer number of lemmas and lengthy rewrites in these papers. Recent LLMs can write correct code for proof assistants, which brings this closer to reality. For example, Anthropic reports that LLMs wrote a 13-million-line Lean proof of Fermat's Last Theorem within 11 days [7].

In quantum computing, proof assistants and, more recently, LLM agents have mostly been applied to programs rather than to circuit equational theories. Mechanized quantum compilers and program

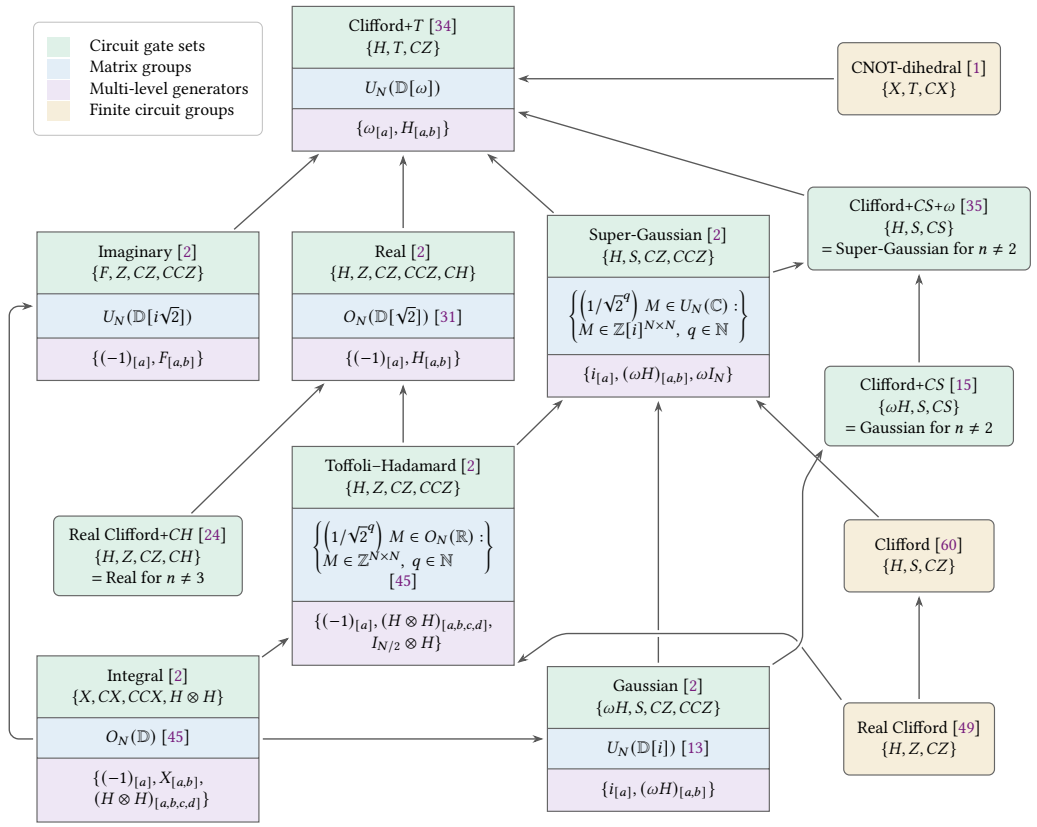


Fig. 2. Restricted Clifford+T circuit families and their underlying matrix groups, with $N = 2^n$ and $n \geq 1$. Each matrix group is the one its circuits represent with at most one clean ancilla. Arrows denote inclusion of ancilla-free circuit groups and, where shown, matrix groups. Mint-only boxes mark when related circuit groups agree; each exceptional inclusion is strict. Here $\mathbb{D} = \mathbb{Z}[1/2]$, $\omega = e^{i\pi/4}$, $F = (H + iI)/\sqrt{2}$, and q ranges over nonnegative integers. Subscripts specify distinct basis levels; $I_{N/2} \otimes H$ and ωI_N act globally. In both Gaussian cases, $X_{[a,b]} = (\omega H)_{[a,b]} i^2_{[b]} (\omega H)^\dagger_{[a,b]}$ is redundant. Clifford+CS+ω denotes adjoining the scalar ω to Clifford+CS.

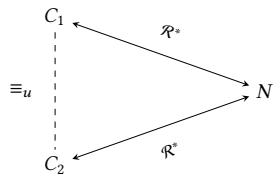
logics establish program correctness or the soundness of circuit transformations [21, 41, 55, 69]. The mechanized completeness results for Π and its square-root extension are coherence arguments [20, 22]. Human-supervised, AI-assisted workflows have also formalized quantum program logics, algorithms, and components of interactive proofs [46, 57, 68]. In concurrent work, Shirobokov et al. [62] build a quantum program library in Lean, where researchers write frozen specifications of correctness and cost and teams of LLM agents synthesize and verify new programs against them.

Orientation. Sections 2 and 3 introduce the proof strategies and the QCert framework. Section 4 presents the formalized results and compares the Agda and Lean formalizations, and Section 5 describes the corrections to the source papers. Section 6 presents a new completeness theorem for four-bit reversible circuits and its formalization. Section 7 discusses limitations and future work. Appendix D works through the multi-qudit Clifford formalization as a case study, and Appendix E proves the theorems of Section 6.

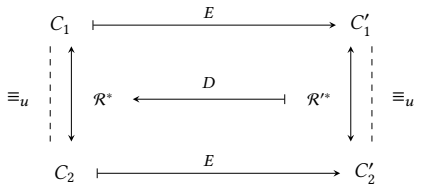
2 Preliminaries

A circuit is a word over an alphabet $\mathcal{G}$ of gates. A rewrite rule is then a relation between words over this alphabet, and a rule set $\mathcal{R}$ generates a congruence on circuits. To a circuit C we associate an *interpretation* $[\![C]\!]$, which is its underlying matrix. The completeness results we formalize have matrix entries in the finitely generated rings $\mathbb{Z}\left[1/\sqrt{2}, i\right]$, $\mathbb{Z}\left[1/2, i\right]$, $\mathbb{Z}\left[1/\sqrt{2}\right]$, and $\mathbb{Z}\left[1/2\right]$, which make them exactly representable in Agda. We leave the formalization of universal circuits taking values in $\mathbb{C}$ for future work.

There are two main strategies for proving completeness of a rule set $\mathcal{R}$ over a gate set $\mathcal{G}$, both of which we support. Consider any two circuits C_1 and C_2 over $\mathcal{G}$ that have the same linear map, $[\![C_1]\!] = [\![C_2]\!]$. We need to show that C_1 can be rewritten into C_2 using $\mathcal{R}$. Figure 3a outlines the first strategy. It specifies a unique normal form for each linear map [1, 12, 17, 32, 33, 43, 44, 49, 58, 60], often by giving an exact synthesis algorithm [13, 23, 31, 36, 45]. Completeness follows by proving that every circuit can be transformed into its normal form using the rewrite rules. By uniqueness, C_1 and C_2 have the same normal form N. Reversing the path from C_2 to N and concatenating it with the path from C_1 to N then gives the desired path of rewrites.



(a) The rewrites from C_1 and C_2 to their common normal form N using $\mathcal{R}$ give rewrites from C_1 to C_2 using $\mathcal{R}$.



(b) Completeness of $\mathcal{R}'$ gives rewrites between $C_1' = E(C_1)$ and $C_2' = E(C_2)$. Translating this proof back gives rewrites between C_1 and C_2 using $\mathcal{R}$.

Fig. 3. Two strategies for proving completeness of $\mathcal{R}$ over a gate set $\mathcal{G}$. $\overset{\mathcal{R}^*}{\longleftrightarrow}$ means applying a finite sequence of rewrites using $\mathcal{R}$ in either direction, where $*$ denotes zero or more steps.

Figure 3b illustrates the second strategy, which uses an already complete rule set $\mathcal{R}'$ over a different gate set $\mathcal{G}'$ [3, 14–16, 18, 24, 26, 27, 30, 59]. Here, E and D translate circuits from $\mathcal{G}$ to $\mathcal{G}'$ and back, replacing each gate with a circuit over the other gate set. Both E and D preserve circuit composition and the underlying linear map. To transfer completeness from $\mathcal{R}'$ to $\mathcal{R}$, we first show that every rule of $\mathcal{R}'$, translated by D, can be derived using $\mathcal{R}$. We can then replace each step of the proof between C_1' and C_2' by its translated derivation, giving $D(C_1') \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_2')$. It then remains to prove that the rewrite $C_j \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_j')$ is derivable, for which it suffices to prove the round-trip rewrites $g \overset{\mathcal{R}^*}{\longleftrightarrow} D(E(g))$ for each $g \in \mathcal{G}$. Sometimes the two strategies are combined, as in the scaled dyadic extension of [45].

The exact synthesis results used by the first strategy present groups of matrices rather than circuits (the first block of Table 1). Their generators are *multi-level* matrices. For an $m \times m$ matrix M and distinct indices $a_1, \ldots, a_m$ in $\{0, \ldots, n-1\}$, the *m-level* matrix $M_{[a_1,\ldots,a_m]}$ is the $n \times n$ matrix that acts as M on the standard basis vectors $e_{a_1}, \ldots, e_{a_m}$ and as the identity on the others. For example, $(-1)_{[a]}$ negates e_a, and $\mathrm{X}_{[a,b]}$ exchanges e_a and e_b. Circuits on k qubits have matrices of dimension $n = 2^k$, but these groups are defined in every dimension n. We introduce multi-level matrices for two reasons. The gap we find in Section 5 is in a presentation by one- and two-level matrices [31], and our Toffoli–Hadamard completeness result builds on a presentation by one-, two-, and four-level matrices [45].

For proofs based on exact synthesis, QCert supplies the common induction argument. Each development proves a Main Lemma connecting the algorithm's reduction steps to the rewrite rules. This isolates the algorithm-specific case analysis—the main work in each formalization—from the reusable completeness proof. Appendix A gives the argument.

Starting from a complete rule set, we can remove a redundant rule by deriving it from the other rules, thereby preserving completeness. To show that the resulting rule set is minimal, we need to give, for each remaining rule R, an interpretation of circuits that every other rule preserves but R does not. Any derivation using the other rules preserves this interpretation, so R cannot be derived from them [18, 25].

3 The QCert Framework

Our Agda framework QCert states every completeness theorem as a group presentation. Gates are indexed by their arity, the number of wires on which they act. Each source relation is stated on its original wires, and shared *structural rules* allow it to be used inside a larger circuit. These rules are independent of the gate set and make explicit the circuit structure that source papers often leave implicit.

We associate to circuits a matrix semantics $[\![-]\!]$ with entries in a finitely generated ring such as $\mathbb{Z}\left[1/\sqrt{2}, i\right]$. This defines a group homomorphism to a unitary matrix group G, whose group structure is checked once per ring using Agda's standard library. QCert extends the interpretation from gates to circuits and proves the structural rules sound once for all gate sets. For soundness, each development then needs to check only its own rules.

Soundness and completeness say the relations present the image of $[\![-]\!]$, which is the subgroup of G generated by the gates. In Agda, mathematical structures such as groups are defined as records. The record $\mathcal{R}$ IsSubPresentationOf G contains the semantics $[\![-]\!]$, with proofs of soundness and completeness. The record $\mathcal{R}$ IsPresentationOf G adds surjectivity, so the relations present all of G.

Figure 4 summarizes the two obligations: checking each rule's matrix equality for soundness, and deriving a known complete rule set $\mathcal{T}$ from $\mathcal{R}$ for completeness.

Each formalization ends with one of these records. For example, Statement (1) states that the real stabilizer relations [49] present the subgroup generated by their gates, while Statement (2) states that the Hadamard-Π relations [31] present the specified orthogonal group. The fields in Statements (3) and (4) state the two directions: equal matrices imply a derivable circuit equation, and every derivable equation preserves the matrix.

```
presentation : ∀ n → (n CRel,_===_) IsSubPresentationOf U (2 ^ n)          (1)
presentation : ∀ n → (n MRel,_===_) IsPresentationOf (O n)                 (2)
complete : ∀ {n} {w v : Circuit n} → [[ w ]] ≈ᵁ [[ v ]] → ⊢ w ≈ v          (3)
sound : ∀ {n} {w v : Circuit n} → ⊢ w ≈ v → [[ w ]] ≈ᵁ [[ v ]]             (4)
```

The minimality statement in Statement (5) says that the rule set is complete on n wires and becomes incomplete there if any rule is removed [18].

```
minimal-complete : ∀ {n} → (n ⊢_===_) IsMinimal × (n ⊢_===_) IsComplete     (5)
```

For each presentation, a reader checks four components against the source paper: the transcribed relations, the target group, the matrix of each gate, and the assignment of $[\![-]\!]$ to the record. Each Agda development collects these components and its main theorem in a `MainTheorem` module. The file `MainTheorems.agda` restates the results; the corresponding Lean statements are in each development's `Theorem` module, which `QCert/Table1.lean` imports.

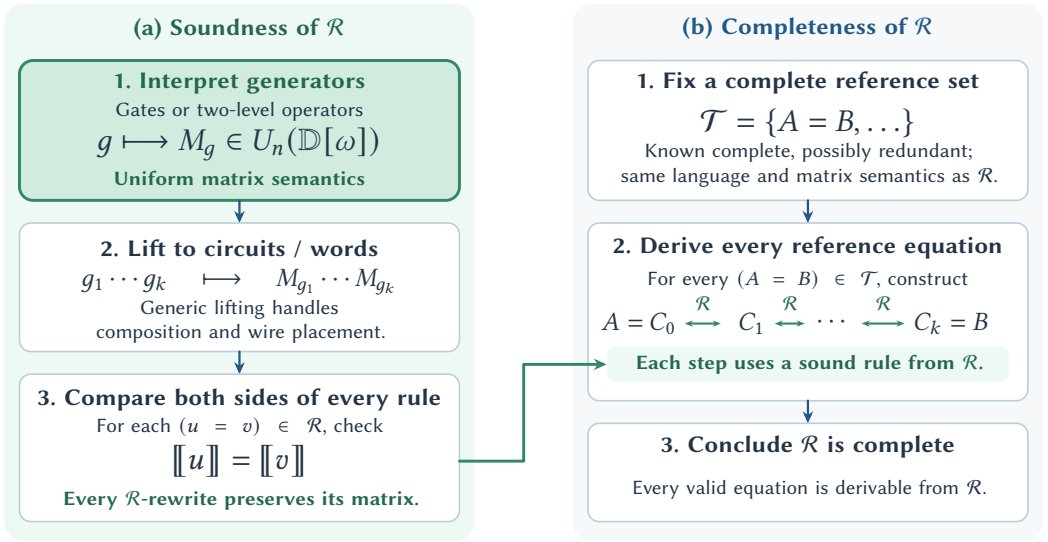


Fig. 4. QCert's shared matrix interpretation supports both proof obligations. The left column checks every rule of $\mathcal{R}$; the right derives every rule of a known complete set $\mathcal{T}$ in the same circuit language and matrix semantics. Each rewrite step uses a rule of $\mathcal{R}$, whose soundness is checked on the left. The displayed $U_n(\mathbb{D}[\omega])$ covers qubit fragments [2], with $\mathbb{D} = \mathbb{Z}[1/2]$ and $\omega = e^{i\pi/4}$. Qudit circuit families use other rings.

Extensions of the Earlier Framework. Besides matrix semantics and minimality, QCert extends the one- and two-wire syntax of Bian and Feng [10] with three-wire gates such as Toffoli and explicit scalar generators for expressing exact equality, including global phase. It also supplies the structural swap laws of PROPs [47], so each paper need only state its own relations. Appendix B gives the encoding conventions, theorem notation, and implementation details.

3.1 Formalization Workflow

Our workflow has three parts. First, before formalizing a paper, we check its rewrite rules, its intermediate results, and each line of its derivations with exact matrices in short scripts. This caught most typos before any Agda code was written. Second, we state each main theorem in QCert before a coding agent starts, so the agent only supplies the proof. Finally, for long derivations, the agent writes a script that turns each line of the paper into one step of the Agda proof, so a failing step points to a specific line of the paper.

Direct evaluation of long matrix products was expensive, so QCert uses a shared exact procedure for matrix equality. We record this technique in the agents' instructions so that later formalizations reuse it; Appendix B.3 gives the implementation.

3.2 Avoiding Completeness Lost in Translation

For a formalization to be useful, its formal statement must say what the source paper claims. We make this connection explicit in three ways.

First, we define the target group independently of the circuit syntax, using matrices or an algebraic construction. A completeness theorem then relates the circuit equations to a semantics that does not depend on them, rather than to a group built from the equations themselves.

Second, when a source rule set must be changed, the formal theorem names the revised set. This lets a reader tell a repair to the proof apart from a correction to the theorem itself (Section 5).

Finally, we verify general completeness arguments and group constructions once, including Reidemeister–Schreier coset enumeration [48] and presentations of semidirect products. Each formalization that uses them only needs to prove their hypotheses and check the connection to its own circuit semantics.

For example, Figure 5 shows how we formalize the multi-qudit Clifford result of Bian, Li et al. [12] using the first and third choices. Its semantic model is the semidirect product of the Pauli phase space by the symplectic group, which we define independently of the circuits. To prove completeness, we combine a normal-form proof for the symplectic group with the framework's semidirect-product and scalar-extension theorems. We give the full case study in Section D.

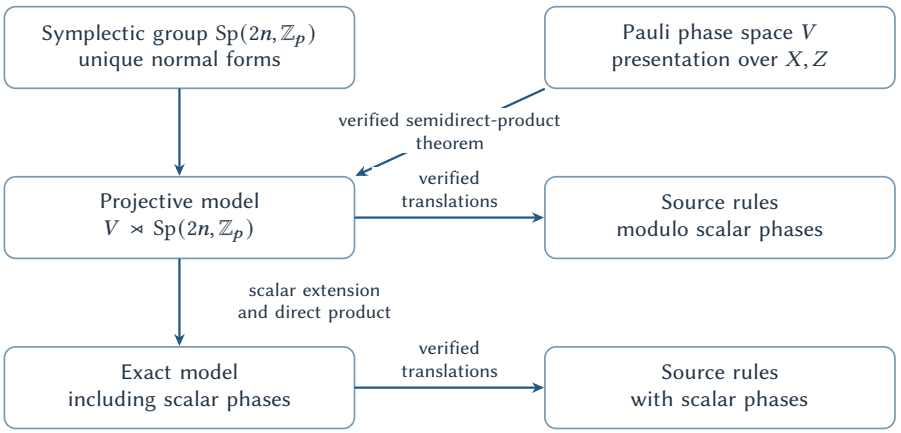


Fig. 5. The formalization of the multi-qudit Clifford result of Bian, Li et al. [12], assembled from verified components. The symplectic normal-form proof and the Pauli presentation combine into a projective model; restoring phases gives an exact model. Arrows denote checked constructions and translations of presentations. Appendix D gives the full case study, including the intermediate rule sets (Appendix D.6) and the scalar corrections (Appendix D.7).

4 Formalized Completeness and Minimality Proofs

The sixteen formalizations in Table 1 contain about 377,000 lines of Agda code and use no postulates, and each theorem is as general as the result in its source paper. Moreover, all matrices are computed exactly. Their entries lie in algebraic number rings, whose elements are given by finitely many integers, so Agda checks automatically whether two matrices are equal. Over the real or complex numbers, no such automatic check exists, which limits the equality checking of QCert to discrete gate sets, as we discuss in Section 7.

Figure 6 adds completeness status to the families and inclusions of Figure 2, using the survey's notation [56]. It separates circuit relations from presentations of the underlying matrix groups. QCert covers the integral, scaled-real, Gaussian, and real matrix presentations, but circuit completeness for Clifford+T and Clifford+CS is still restricted to small widths. A presentation of the ambient matrix group does not by itself provide relations for every circuit subgroup.

The green checks in Figure 6 correspond to the qubit part of Table 1; the qudit results and minimality proofs are also listed in the table. The asterisks connect this coverage to the source corrections in Section 5.

Source	Object	Result	Agda Mod	Agda time (s)	Lean time (s)
Groups of m-level matrices					
Greylyn [2014]	$U_n(\mathbb{Z}[1/\sqrt{2}, i])$	presentation, $n \leq 4$	34	37.28	145.34
Bian and Selinger [2021]	$U_n(\mathbb{Z}[1/2, i])$	presentation	39	53.73	126.99
Li et al. [2021]	$O_n(\mathbb{Z}[1/2])$, L_n	presentations	71	84.52	185.32
Fang et al. [2026]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	46	48.33	231.57
Clément [2019]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	43	67.72	51.99
Circuits					
Selinger [2015]	Clifford	sub-presentation	32	35.01	88.24
Bian, Li et al. [2026b]	Clifford, odd prime p	presentation of a model	216	499.37	197.32
Makary et al. [2021]	real stabilizer	sub-presentation	31	37.94	79.75
Amy et al. [2018]	CNOT-dihedral	sub-presentation	31	21.24	101.22
Blake [2026b]	phase-affine, prime d	sub-presentations	57	83.30	257.24
Bian and Selinger [2023a]	two-qubit Clifford+T	sub-presentation in U_4	63	135.79	356.79
Bian and Selinger [2023b]	three-qubit Clifford+CS	sub-presentation in U_8	125	298.29	1330.89
Clément [2026]	real Clifford+CH	sub-presentation	348	419.99	5478.79*
Amy et al. [2024]	three-qubit Toffoli–Hadamard	presentation of L_8	203	111.17	134.91
Li et al. [2025]	qutrit Clifford	presentation	90	285.58	179.60
Minimality					
Blake [2026c]	six fragments	minimal, complete	80	180.10	358.87
Total			1,509	2,399.36	9,304.83

Table 1. Completeness and minimality results formalized in QCert. A presentation gives generators and relations for a whole group, and a sub-presentation for the subgroup generated by the gate set named in the Object column of the Circuits block. L_n is the group of orthogonal matrices $N/\sqrt{2}^j$, where N is an integer matrix and $j \geq 0$. Mod counts Agda modules. The Agda times are typechecking times from scratch with Agda 2.8, the times its profiler reports for the row's modules; the Lean times are elapsed folder-build times. Both were measured on the same machine, under the conditions in Appendix C. Both columns exclude the shared library and other developments on which a row depends. The Result column describes the Agda statements; Appendix C explains the semantic differences. *Includes two unused legacy helpers that failed; the main theorem chain passed.

Two qualifications apply to the Agda results in Table 1. The two-qubit Clifford+T and three-qubit Clifford+CS developments [14, 15] reuse the authors' Agda proofs and connect them to QCert's shared matrix presentations (Section 3). The qudit Clifford theorem of Bian, Li et al. [12] presents an independently constructed algebraic model; Appendix D.3 explains its connection to the intended unitary semantics.

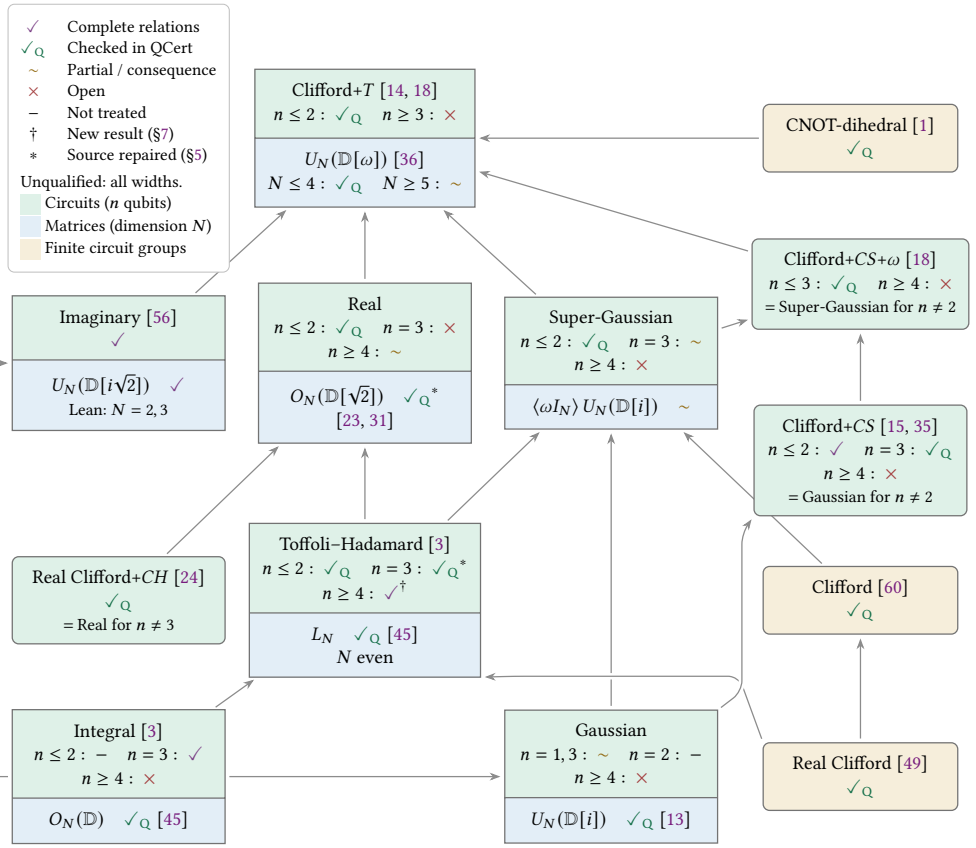


Fig. 6. Completeness progress on the same families and inclusion arrows as Figure 2. The dagger marks the new Toffoli–Hadamard result, whose exposition is in preparation. Gates and multi-level generators are omitted; upper compartments concern ancilla-free circuits, lower compartments matrix presentations. Here n counts qubits and N is matrix dimension ($N = 2^n$ for a circuit). The survey's ~ includes consequences whose gate-set transfer is not formalized in the corpus; a plain check does not assert QCert verification. Imaginary-family checks refer to new computer-assisted proofs [56]; their complete Lean checks cover only $N = 2, 3$ and are outside this corpus. The higher-dimensional Clifford+T matrix presentation remains incomplete. L_N is defined in Table 1; the other notation is as in Figure 2.

4.1 Comparison of the Agda and Lean Formalizations

The Lean [29] version formalizes the same sixteen results using Mathlib's standard definitions [64], with matrix groups over real or complex numbers. It translates the Agda framework, so agreement between the two does not independently validate their shared design. Figure 7 compares proof size, verification time, and execution time. Lean developments generally use fewer lines and Agda records lower verification times in thirteen rows, but proofs and measurement methods differ. Translating the same proof, for Toffoli–Hadamard and for the four-bit circuits of Section 6, yields similar amounts of text, and Lean checks the translation in 3.3 to 3.4 times Agda's time. Agda provides executable normal forms, whereas Lean's are existence results or `noncomputable` definitions. Panel (c) measures their executable circuit-equivalence procedures, where Lean is faster

from three qubits onward. Appendix C gives the detailed comparison and measurement methods, summarized in Table 2.

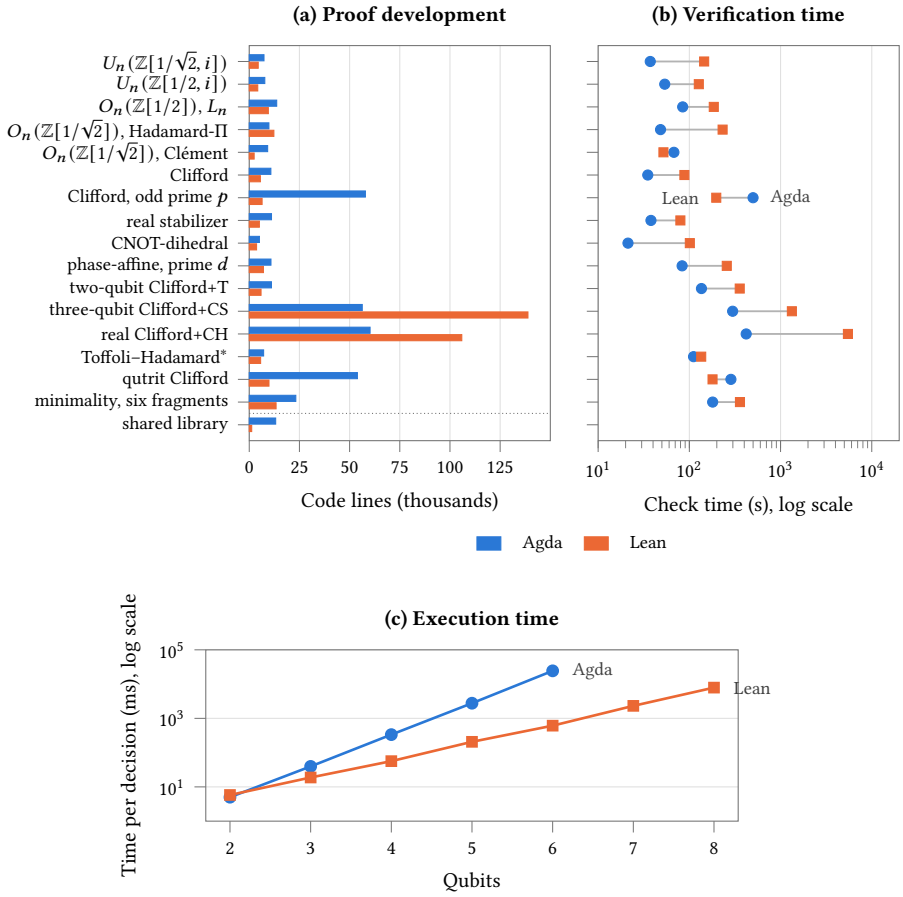


Fig. 7. The Agda and Lean formalizations of each row of Table 1. (a) Code lines, excluding blank lines and comments. For Toffoli–Hadamard* it shows another pair, one proof in both assistants (Appendix C): a second Agda proof and its translation to Lean. The last row is the shared library of each framework. (b) The check times of Table 1, Agda (circles) and Lean (squares), measured on the same machine, with methods described in Appendix C. (c) The time each compiled formalization takes to decide that two Clifford circuits of 100 gates have the same matrix, by the procedure its soundness proof uses, as medians over three runs of five random circuits.

5 Corrections to the Source Papers

The formalizations in Table 1 identify two kinds of issues in the source papers. Most are typos, and some of them appear in statements that the proofs rely on. These are easy to fix. Two others are harder to repair. One is a gap in a proof, and the other is a relation missing from a theorem. We describe both below.

The gap is in the proof of Theorem 4.14 of Hadamard-Π [31], which states that the relations in its Figure 6, over the one- and two-level matrices $(-1)_{[a]}$, $X_{[a,b]}$, and $H_{[a,b]}$, present the group

$O_n\left(\mathbb{Z}\left[1/\sqrt{2}\right]\right)$. One case of the proof relies on Lemmas A.16 and A.17, which have counterexamples in dimension 6, and the case analysis built on them misses a matrix in dimension 9. Their main result is however still correct: Clément's unpublished notes [23] present the same group with different relations, and his case analysis covers the missing case, using which the formalization of Theorem 4.14 is completed. We also prove that the two sets of relations are equivalent in every dimension and formalize Clément's proof [5].

The relation is missing from Theorem 5.2 of [3]. It states that the relations in its Figures 4 and 7, over the Toffoli and $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ gates, present the group $O_8(\mathbb{Z}[1/2])$. This is the first of two steps towards presenting the scaled dyadic orthogonal group L_8 by three-qubit Toffoli–Hadamard circuits. The relation r_{K} in its Section 5.5, which splits K on two qubits into two copies controlled by the third qubit, is in neither figure. Nor can it be derived from them. Hence, Theorem 5.2 fails as stated. Our formalization [6] shows that r_{K} is independent by giving an interpretation of circuits that every other relation preserves but r_{K} does not, and proves the presentations of $O_8(\mathbb{Z}[1/2])$ and L_8 with r_{K} restored.

6 A New Result: Complete Equations for Four-Bit Reversible Circuits

Beyond the published results, we use QCert to prove and check a new completeness theorem for classical reversible circuits. The gates X (NOT), CX (CNOT), and CCX (Toffoli) generate the reversible circuits on bits. Without ancillas, circuits on $k \leq 3$ wires realize every permutation of the 2^k bit strings, while on $k \geq 4$ wires every gate is an even permutation and the circuits realize exactly the even ones [61]. Complete equational theories are known for linear reversible circuits [43, 58] and for circuits over multiple-control Toffoli gates [32, 33]. We take the gate set $\{\mathrm{X}, \mathrm{CX}, \mathrm{CCX}\}$ on at most four wires, and the theory $\mathcal{T}$ of the seven equations in Figure 8. Equations (T1)–(T6) act on at most three wires and (T7) on four. Two circuits are *equal modulo* $\mathcal{T}$ when one can be rewritten into the other by these equations and the structural rules of Figure 9, every equation used in either direction.

Write G_k for the circuits on k wires modulo $\mathcal{T}$. Each gate is an involution modulo $\mathcal{T}$, so G_k is a group, and the semantics $[\![-]\!]$, which maps a circuit to its permutation of the bit strings, is a group homomorphism from G_k.

Theorem 6.1 (completeness). *For $k \leq 4$, two circuits on k wires over $\{\mathrm{X}, \mathrm{CX}, \mathrm{CCX}\}$ are equal modulo $\mathcal{T}$ if and only if they denote the same permutation. Hence $G_k \cong \mathrm{Sym}(2^k)$ for $k \leq 3$ and $G_4 \cong \mathrm{Alt}(16)$.*

The theorem also has a constructive form. For an eight-element set S of the sixteen states, write r_S for a fixed circuit that maps S onto the states whose first bit is 0. For $B \in \mathrm{Sym}(8)$, write $\mathrm{D}(B)$ for a fixed circuit that applies B to wires 2–4, and for $E \in \mathrm{Alt}(8)$, write $\mathrm{C}(E)$ for one that applies E to wires 2–4 when wire 1 is 1 and never targets wire 1. The *block normal form* $\mathrm{NF}(E, B, S)$ first applies r_S, then $\mathrm{D}(B)$, and then $\mathrm{C}(E)$.

Theorem 6.2 (block normal form). *Every four-wire circuit is equal modulo $\mathcal{T}$ to exactly one block normal form, and $(E, B, S) \mapsto [\![\mathrm{NF}(E, B, S)]\!]$ is a bijection from $\mathrm{Alt}(8) \times \mathrm{Sym}(8) \times \binom{\mathbb{F}_2^4}{8}$ onto $\mathrm{Alt}(16)$. For every circuit w, the circuit that applies w and then $\mathrm{NF}(E, B, S)$ is equal modulo $\mathcal{T}$ to $\mathrm{NF}(E', B', w^{-1}S)$, where E' and B' are given by an explicit update formula (Appendix E.5).*

Both theorems are proved in Agda [4], and the proof is translated to Lean module for module (Appendix C); Statements (6) to (9) are the four-wire statements of the Agda MainTheorem module, which also states completeness and the presentations of $\mathrm{Sym}(2^k)$ on fewer wires. Here $[\![w]\!]$ is the function of w on bit vectors, $\doteq$ is pointwise equality, and $u \bullet w$ applies u and then w.

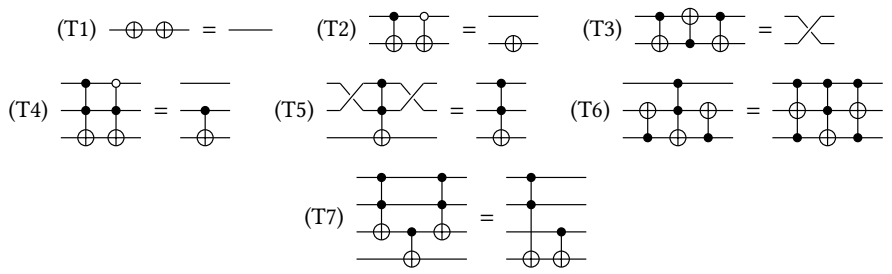


Fig. 8. The theory $\mathcal{T}$ for circuits over {X, CX, CCX}. Time runs left to right and wire 1 is on top. A filled control tests 1, an open control tests 0 (it is the filled one conjugated by X on its wire), $\oplus$ is a NOT target, and crossed wires are the structural swap. A CNOT or Toffoli drawn in another position, as in (T3), (T6), and (T7), is the gate on adjacent wires with its target below its controls, conjugated by swaps. Each equation can be used in either direction and, by the structural rules of Figure 9, on any wires of a circuit, adjacent or not and in any order.

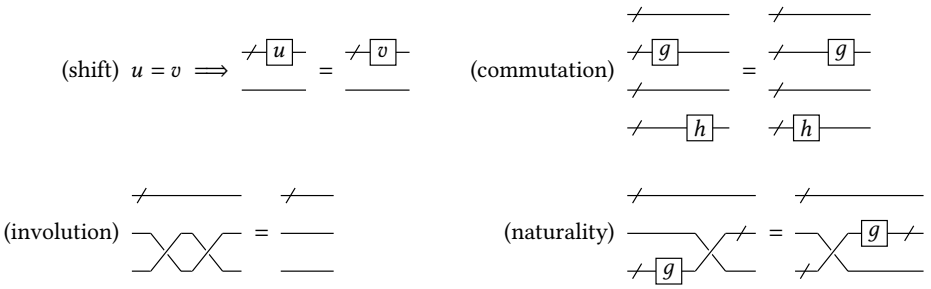


Fig. 9. The structural rules for circuits over {X, CX, CCX}, which QCert supplies for every gate set with a structural swap (Appendix B.1). Here u and v are circuits, g and h are X, CX, CCX, or the swap, and a slashed wire carries the wires of the box on it or, if it has none, any number of idle wires. An equation also holds with an idle wire below it, a gate commutes with any gate above it, the swap is an involution, and a gate slides past a wire that crosses all of its wires; for g the swap, (naturality) is the braid relation. (shift) adds wires only below, while the other rules hold under any idle wires.

```
complete : {w v : Circuit 4} → [[ w ]] ≐ [[ v ]] → ⊢ w ≈ v                    (6)
presentation₄ : (4 CRel,_===_) IsPresentationOf Alt-group 16                  (7)
normal-form : (u : Circuit 4) → ⊢ u ≈ NF (coords u)                           (8)
absorption : (c : Coords) (w : Circuit 4) → ⊢ w • NF c ≈ NF (absorb c w)       (9)
```

Proof idea. A direct proof follows the first strategy of Figure 3, showing that appending a gate to a normal form gives a circuit that rewrites to a normal form. The exposition we are preparing proves the theorems this way, with finite certificates for the 360,360 transitions of the 12,870 sets S under the 28 placed gates and for the subgroup equalities they rely on, about 380,000 deductions in all. The formalization instead combines the second strategy with Reidemeister–Schreier presentations of permutation groups (Section 3.2), so that only small facts are decided by evaluation. It has four steps.

(1) *At most three wires.* Ordered along the Gray code 000, 001, 011, 010, 110, 111, 101, 100, consecutive states differ in one bit, so they are exchanged by a Toffoli gate with suitable control

polarities. Equations (T1)–(T6) derive the Coxeter relations of Sym(8) for these gates. Translating the Coxeter presentation [28] back into circuits gives completeness on three wires, and likewise on fewer.

(2) *Reduction to circuits that keep a wire.* The Reidemeister–Schreier method presents Alt(16) by 28 generators, each a product of two adjacent transpositions, and 238 relations derived from the Coxeter relations of Sym(16). Along the four-bit Gray code, each generator is realized by a circuit of at most eight gates that targets only two wires, so every relation is an equation between circuits that never target some wire. Completeness therefore follows once circuits that never target a given wire, which form a subgroup K, are *faithful*, that is, determined by their permutations.

(3) *The subgroup K.* A circuit that never targets wire 1 preserves its value, so its permutation is block diagonal. Semantically, K is the semidirect product of the circuits on wires 2–4, which are faithful by the first step, and the controlled even permutations $C \cong \mathrm{Alt}(8)$. The group C is generated by the 105 *controlled perfect matchings*, the conjugates of $\mathrm{CX}_{1\to 4}$ by circuits on wires 2–4. We show that each depends only on its matching, and that they satisfy three relation schemas: each is an involution, each conjugates another to the controlled conjugate matching, and two commuting ones multiply to their product when it is a matching. The side conditions of the schemas are decided on the matchings. With the schemas, each of the 54 Reidemeister–Schreier relations of Alt(8) is checked by a certificate of rewriting moves on words of matchings, 412 moves in all.

(4) *The block normal form.* Given completeness, a circuit is equal to the normal form of the coordinates read off its permutation. The update formula follows because the permutation that applies $r_{S'}^{-1}$, then w, and then r_S preserves the halves, where $S' = w^{-1}(S)$.

The only genuinely four-wire equations, which apply (T7), are found by a search that also uses three-wire steps justified by the first step. They include the equation by which a CNOT onto a control of a Toffoli conjugates it into the product of two Toffolis, and three of the five base cases of the conjugation schema, one for each way two matchings can intersect; the longest takes 496 steps. The formalization has 40 modules and about 13,000 lines. Given the shared library, Agda checks it in 85 seconds, and Lean checks its translation in 277. Appendix E gives the details.

7 Discussion and Future Work

We formalize the results of sixteen papers in QCert, using consistent circuit syntax, a unified definition of group presentations, and shared libraries for rings and their matrix groups. Since all formalizations use these definitions, presentations of the same group from different papers can be compared and proved equivalent. The same libraries can be reused for new algorithms, and the induction that turns an exact synthesis algorithm into a completeness proof carries over with little change. The main part specific to each algorithm is the case analysis over matrices and generators. As the most error-prone part of the proof, it is where machine checking in QCert matters most.

Reusing this infrastructure, we have obtained complete equational theories for circuits over $\{X, CX, CCX\}$ and for Toffoli–Hadamard circuits. We are now preparing mathematical expositions of these results.

At the moment, not every completeness result fits into QCert, as its scope is limited in two ways. First, it supports only discrete gate sets, so it does not cover completeness results for circuits with continuous parameters, such as rotations by arbitrary angles [25–27]. Second, control is not a circuit constructor in QCert, so it does not cover results that treat control as a primitive operation on circuits [16, 30, 59].

We plan to extend the infrastructure of QCert to more families of quantum circuits, including the two above. Completeness is still open for many discrete gate sets in higher-dimensional quantum systems, and their case analyses are often too large to check by hand. In particular, we plan to study the completeness of Clifford circuits in all dimensions with the help of QCert. Beyond completeness, the exact synthesis libraries can be reused for T-count optimization of multi-qubit Clifford+T circuits.

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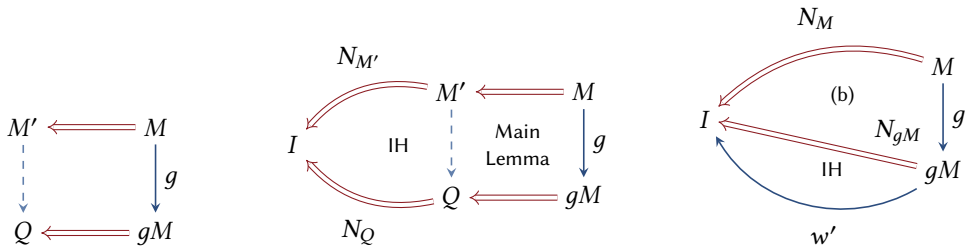
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A Completeness from Exact Synthesis

For the matrix groups discussed in Section 2, normal forms come from exact synthesis, and the proof that every circuit can be transformed into its normal form has the same structure for each algorithm. Each matrix, or *state*, has a *level*, built from quantities such as the least denominator exponent of a column, and the algorithm reduces each state M to the identity in steps that lower the level. These steps form the normal word N_M.

The *Main Lemma* says that for every state M and generator g, a diagram of relations closes the square formed by g and the algorithm's next steps from M and gM, passing only through states of lower level (Figure 10a).

By well-founded induction on the level, the relations then rewrite $N_{gM}\, g$ into N_M (Figure 10b), and by induction on length, every word that reduces M to the identity rewrites into N_M (Figure 10c). Two words with the same matrix therefore rewrite into each other. These inductions have the same form for every presentation, so the main work in each formalization is its Main Lemma.



(a) The Main Lemma. (b) Induction on the level. (c) Induction on length.

Fig. 10. Completeness from the Main Lemma. Plain arrows are words, double arrows are steps or runs of the algorithm, and dashed arrows are words through states of lower level.

B QCert Implementation Details

This appendix gives the circuit encoding, theorem notation, and matrix equality procedure used by the framework in Section 3.

B.1 Wire Placement and Structural Rules

Wires are numbered from 0 at the bottom, and a gate acts on the bottom wires of a circuit. A shift places a circuit on n wires onto wires $1, \ldots, n$ of a circuit with $n + 1$ wires. In the matrix semantics, both placing a gate in a larger circuit and shifting a circuit are Kronecker products with an identity. This defines wire placement once for all gate sets.

Each source relation is stated on as many wires as the paper draws it on. Structural rules allow an equation to be shifted by one wire, gates on disjoint wires to commute, and a scalar to be placed on any wire. QCert extends the earlier framework's one- and two-wire syntax with three-wire gates and their structural rule. Scalars are zero-wire generators; their placement rule allows exact equations to account for global phase.

In the earlier framework, swap is an ordinary gate whose laws must be included in each rule set. QCert also supports a PROP presentation, in which a designated two-wire gate is the swap and the framework supplies its involution law and the laws for moving gates across it. The source rule set then contains only its own relations, each on its original wires; swap and its laws are supplied by the framework. We use this representation for the circuits of [1, 17, 18], and for the reversible

circuits of Section 6, whose structural rules are drawn in Figure 9. That section numbers wires from 1 at the top, as its figures do, so on n wires its wire w is wire $n - w$ here.

B.2 Theorem Notation

The signatures in Section 3 use `n CRel,_===_` for relations on circuits with n wires and `n MRel,_===_` for relations on words over multi-level matrices of dimension n. In Statement (1), the former contains R1–R16 of Figure 1 of [49], lifted to n wires. In Statement (2), the latter contains the relations of Figure 6 of Hadamard-Π [31].

The judgement `⊢ w ≈ v` states that w and v are equal by the relations. The qubit developments interpret matrices in $\mathbb{D}[\omega]$ and use `_≈ᵁ_` for their equality; the qudit developments compare interpretations in their own groups. The `IsSubPresentationOf` record contains the semantics and its soundness and completeness proofs, and `IsPresentationOf` adds surjectivity.

Minimality is stated for the rule set `n ⊢_===_` in Statement (5). Unlike the generated equivalence relation, this retains the individual rules, so removing one is defined. The predicate `IsMinimal` therefore takes this rule set rather than its generated equivalence relation.

B.3 Matrix Equality Checking

The main computational cost came from Agda's conversion checker. It accepts syntactically identical terms immediately; otherwise, it evaluates them to compare their results. For long matrix products, this evaluation can take exponential time. We keep circuit interpretations abstract and decide matrix equality by computing a single Boolean using row operations. For these products, this is orders of magnitude faster than comparing matrices entry by entry. These techniques are included in the agents' instructions so that later formalizations avoid the same slowdowns.

C Details of the Agda and Lean Comparison

The Lean [29] version of QCert formalizes the same sixteen results on top of Mathlib [64]. Once the Agda formalization exists, translating it to Lean is comparatively easy, so the Lean version is not a second achievement of the same kind. Its point is its theorem statements. They rest on Mathlib's standard definitions, with matrix groups over Mathlib's real or complex numbers in place of our exact rings, so a reader can check them against the source papers using definitions that are already widely used and reviewed. For qudit Clifford, the Lean statement also removes an argument that the Agda statement leaves to external mathematics (Section D.3).

Nor is the Lean framework an independent implementation. We instructed the coding agent that wrote it not to consult our Agda implementation, but the Lean framework is nonetheless a translation of the Agda one. Its words, circuit generators, structural rules, and presentation and sub-presentation predicates are those of the Agda framework, down to the names of the definitions and the wording of their documentation. The two formalizations therefore share the design of their framework, and would share any mistake in it, so their agreement is weaker evidence than two independent formalizations would give. They use different representations and sometimes different proofs, so the theorem statements of each must be checked against the source papers separately. We compare their definitions, proof development, verification time, and execution time; Table 2 summarizes the comparison. These observations concern our two formalizations, not Agda and Lean in general.

Definitions and Semantics. The translation keeps the definitions but adapts their representation to Mathlib. Agda uses its own word type, with explicit brackets, and the setoid-based algebra of the standard library. A presentation is a record containing a semantics and a group isomorphism, or a monomorphism for a sub-presentation. Lean uses Mathlib's free monoid and the monoid

	Agda	Lean
Proof development		
Logic	constructive: every module `--safe`, no postulates	classical: every main theorem uses `Classical.choice`
Library	standard library (104 modules imported), EucDomain	Mathlib (56 modules imported directly)
Proofs	terms, by pattern matching and equational reasoning; reflection-based solvers	tactics: about 75,000 tactic blocks, 69,000 automation calls
Reused from the sources	the authors' Agda proofs for Clifford+T and Clifford+CS	no proofs; the authors' coset-action data for Clifford+T
Modules	1,509	795
Code lines	376,627	338,866
Verification time		
Decided equalities	a Boolean reduced to `true`	`decide +kernel`
Check time, sixteen rows	2,399 s	9,305 s
Faster for	thirteen rows	three rows
Execution time		
Normal forms	executable functions	existence theorems or `noncomputable` definitions (460 declared)
Benchmarked procedure	matrix of each circuit, by products of stored matrices	simulation of each circuit on every basis vector
Compiled through	GHC (Haskell)	C
Per decision, Clifford, 100 gates	5.0 ms on 2 qubits, 24 s on 6	5.8 ms on 2 qubits, 0.61 s on 6

Table 2. The Agda and Lean formalizations compared. Counts are over the files under version control in the sixteen development folders of Table 1, without the shared libraries, and code lines exclude blank lines and comments. Check times are those of Table 1, and execution times those of Figure 7(c).

congruence generated by the relations. Its presentations are predicates in `Prop`, from which it derives isomorphisms out of the quotient. In both, the circuit infrastructure, the structural rules, and general completeness arguments are shared between developments.

The semantic targets also differ. Most Agda developments work directly with exact representations of matrix entries. Lean states its matrix groups over Mathlib's real or complex numbers and proves bridges to them from computable exact rings. In particular, the Agda qudit Clifford development presents the constructed model of Appendix D, whereas Lean proves soundness and completeness against complex matrices for every odd prime. A common source therefore does not imply identical statements or intermediate lemmas.

Proof Development. Excluding blank lines and comments, the Agda formalization of Table 1 has about 377,000 lines and the Lean formalization about 339,000 (Table 2). Figure 7(a) compares them row by row, except that for Toffoli–Hadamard it shows another pair, one proof in both assistants

(see below). Lean's is shorter in thirteen of the sixteen rows, by a factor of five or more for qudit Clifford and qutrit Clifford, where the two take different routes. It is longer for Hadamard-Π, three-qubit Clifford+CS, and real Clifford+CH. Both counts include the derivations and certificates that scripts generate. The Agda proofs are explicit terms, built by pattern matching and equational reasoning, with reflection-based solvers for associativity and ring identities. Its library develops the group theory the proofs need, in about 13,200 lines against 1,200 in Lean. The Lean proofs rely on Mathlib's algebra and on tactics, with about 69,000 calls to automation such as `simp`, `omega`, `decide`, and `ring`.

Both implementations reuse normal-form, exact-synthesis, and translation arguments. Both also check generated derivations with a replay procedure that is proved correct. The proof assistant checks the computation and the theorem that a successful replay gives a derivation, so the scripts that search for derivations need not be trusted. For Clifford+CS, Lean constructs its own coset certificates. The trust checks reflect the different foundations. Every Agda module is `--safe` and uses no postulates, so the Agda corpus is constructive. The Lean corpus uses no `native_decide`, and its main theorems depend on no unfinished proof. An audit of 105 declarations reports only the axioms `propext`, `Classical.choice`, and `Quot.sound`.

Verification Time. Table 1 and Figure 7(b) give the check times, both measured on the same machine. In total Agda took 2,399 seconds and Lean 9,305, or 1,979 and 3,826 without real Clifford+CH, whose Lean build alone took 5,479 seconds, more than half of the total. Lean took between 1.2 and 4.8 times as long as Agda on twelve rows, and 13 times as long on real Clifford+CH. Its five-wire certificate modules `Family267` and `Family268` alone took about 1,741 and 1,369 seconds. Lean was faster for three rows, Clément's presentation, qudit Clifford, and qutrit Clifford, whose Lean proofs take different routes. Both assistants decide equalities of matrices and certificates by evaluation in the checker. Agda reduces a Boolean to `true`, and Lean uses `decide +kernel`.

Both columns were measured under WSL on an Intel Core i5-13400TEF with sixteen logical CPUs, about 26 GiB of RAM, and 64 GiB of swap. The Agda column comes from a check from scratch on 8 November 2026, with Agda 2.8.0 and version 2.4 of its standard library. We removed the repository's interface files, kept those of the standard library and EucDomain, and checked the shared library and then each row in an Agda process of its own, through a module that imports every module of the row. Agda's profiler reports the time spent on each module without its imports, and a row's time is the sum over its own modules, wherever they were checked. It leaves out process startup and the loading of interface files: the whole check, shared library included, took 3,210 seconds of wall time, and no process used more than 5.1 GiB. No other proof assistant ran during the check. The Lean column comes from a clean build on 5–6 November 2026, with Lean 4.30.0 and the pinned Mathlib revision. We removed QCert's build artifacts, kept the Mathlib and other dependency caches, and built the folders one at a time in dependency order with `LEAN_NUM_THREADS=2`. Each time is the elapsed wall time of a folder, including Lake startup and artifact generation, and a shared dependency is counted only in the folder that defines it. The two columns therefore measure different things, per-module checking time in Agda and folder build time in Lean. Proof constructions, module boundaries, parallelism, and artifact generation also differ, so the differences cannot be attributed to the proof assistants alone. We have yet to separate the time Lean spends running automation from the time it spends checking the resulting proofs.

The build covered all 810 tracked library modules, excluding three generator templates, and took **9,415.37 seconds (2 h 36 min 55 s)** in timed build commands. Of these modules, 808 passed. Two unused legacy helpers in real Clifford+CH failed, `ItemBFun` at `omega` proofs and `ItemASteps` at the heartbeat limit. Neither is imported by the main theorem chain or the audit, both of which passed, and the real Clifford+CH entry of Table 1 includes the whole attempted build. Beyond

the sixteen development folders, the shared `Circuit`, `Exact`, `Matrix`, and `NumberTheory` folders took 17.91, 15.51, 6.88, and 15.29 seconds, and the five root modules, including the audit, took 54.95 seconds. Lean builds of another project ran on the same machine during part of the run. We therefore re-timed the twenty folders and root modules they overlapped and report the second timing, during which the other project used about a quarter of one CPU on average. The two runs total 9,604 and 9,415 seconds, and the two timings of each development folder agree within 15%, except for Toffoli–Hadamard (178 and 135 seconds).

One Proof in Both Assistants. For Toffoli–Hadamard, the formalizations of Table 1 have 28,629 lines in Agda and 4,401 in Lean, and they prove the result by different routes. The Agda proof replays the derivations in the authors' own files as generated certificates, which make up 22,572 of its lines. The Lean proof was written from the paper, over an alphabet of macro gates, with a Knuth–Bendix system for the symmetric group S_8 that the kernel checks. To compare the two assistants on one proof, Figure 7(a) shows another pair instead (∗). We wrote a second Agda proof of the result, which finds its circuit derivations by a breadth-first search over the relations of Figure 4 of [3], and a coding agent translated it to Lean module for module, keeping its definitions, derivations, and proof route; this time we told the agent to follow the Agda. A coding agent also began translating the first Agda proof in the same way, porting its scripts so that they write the certificates in Lean, and we stopped it after 46 of the 203 modules: the relations, the 218 derivations replayed from the authors' files, the 642 certificates of the permutation fragment, and the frames built on them, 17,862 of the Agda proof's lines. On one proof the two assistants need about the same amount of text: 182,301 non-blank characters outside comments in Agda and 191,589 in Lean for the second proof and its translation, and 1,056,950 and 1,030,864 for the translated modules of the first. In lines the translation of the second proof is shorter, 5,672 against 7,161, only because its generated derivations put several rewrite steps on a line where Agda's put one, while the translated modules of the first, whose certificates take a step per line in both, are 6% longer, 18,946 against 17,862. The hand-written part of the second proof's translation is the larger of its pair, mostly because it proves the facts about S_8 that the Agda proof imports from the shared library. Lean took 164 seconds to check it and Agda 48, 3.4 times as long, while on the two proofs of Table 1, timed in the same way, Lean took 118 seconds and Agda 111, 1.1 times as long (Table 1, timing Lean's whole build, has 135 and 111). So the pair that Table 1 and Figure 7(b) report reflects the difference between the proofs more than the difference between the assistants.

The new result of Section 6 gives a third pair. Coding agents translated its 40 Agda modules to Lean module for module, again told to follow the Agda, and ported the scripts that generate its derivations, tables, and certificates so that they write Lean. Seven further Lean modules translate what the Agda imports from the shared library: the Coxeter presentation of the symmetric groups, signs and the alternating groups, the swap rules, and the rewriting engine. Again the two need about the same amount of text: 421,726 non-blank characters outside comments in Agda and 467,667 in Lean, or 431,042 without the seven library modules, in 13,135 and 12,177 lines. Lean took 277 seconds to check the translation and Agda 85, 3.3 times as long, close to the ratio of the Toffoli–Hadamard pair. Agda checks the 496-step derivation of a base commutation by one evaluation, whereas one kernel check of it in Lean had not finished after eight minutes, at 11 GB of memory, so the translation cuts each long derivation into segments of at most 40 steps. These six times were measured on 10 November 2026 on the machine of Table 1, by its method, with no other proof assistant running.

Execution Time. The Agda normal forms are executable functions. The Lean developments obtain normal forms either as existence theorems or as `noncomputable` definitions. For example, the

Lean proofs for the Clifford and real stabilizer circuits of [49, 60] show that every circuit equals a normal form without defining a function that computes it, and the exact synthesis of the two-level groups is noncomputable. In all, 460 definitions in the sixteen Lean developments are declared `noncomputable`, so there is no Lean normalizer to time. Both formalizations can, however, execute the procedure that decides whether two circuits have the same matrix, which their soundness proofs run. We compiled this procedure for the Clifford circuits of [60] to native code, through Agda's GHC backend and Lean's C backend. We then timed both on an Intel Core i5-14400, on the same random circuits of 100 gates over H, S, and CZ, deciding that each circuit w has the matrix of $w\,S^4$ (Figure 7(c)). On two qubits Agda is slightly faster, at 5.0 against 5.8 milliseconds per decision. From three qubits on Lean is faster, by a factor of 2.1 on three qubits and 40 on six, where Agda takes 24 seconds and Lean 0.61. The gap comes mostly from the procedures rather than the compilers. Agda computes each circuit's matrix as a product of stored $2^n \times 2^n$ matrices, so its work grows about eightfold with each qubit. Lean simulates the circuit on each basis vector, so its work grows about fourfold.

D Case Study: Multi-Qudit Clifford Completeness

The multi-qudit Clifford result of Bian, Li et al. [12] shows how QCert turns a circuit completeness proof into reusable, checked components. Here each qudit has odd prime dimension p; the result covers every number n of qudits and every such p. Its sixteen rules must derive every valid equation between Clifford circuits. It illustrates two design choices of Section 3.2, defining the semantic group independently of the circuit syntax and assembling completeness proofs from general theorems about groups, as Figure 5 summarizes.

D.1 From Circuit Equations to Checked Derivations

The circuits use the Fourier gate H, the phase gate S, and the controlled-phase gate CZ. We first work projectively, identifying operators that differ only by a global phase. Figure 11 gives a small example of the translation into QCert: a swap is a word in the generators, its laws are circuit equations, and a derived equality records the rule used at each step. The framework supplies the structural rules that let the same derivation apply inside a larger circuit. The corresponding Agda definitions and proof terms are in Appendix D.4.

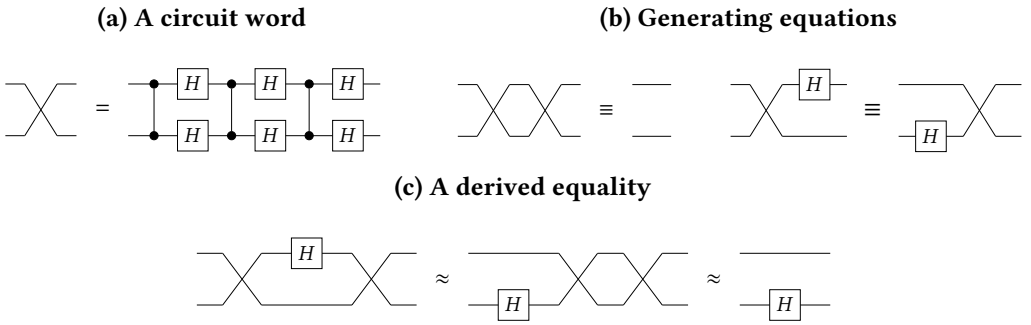


Fig. 11. Circuit syntax, rules, and reasoning in the projective Clifford formalization. Crossed wires denote swap, and connected dots denote CZ. Diagrams are read from right to left. In (c), moving H across one swap and cancelling the two swaps moves H to the other wire. The generating equations (≡) justify the derived equivalence (≈).

To prove completeness, we must also specify what a circuit means. We first consider its action on Pauli operators, ignoring phase factors. Each wire contributes a pair (a, b) modulo p, describing $X^a Z^b$ for the generalized Pauli shift and phase gates X and Z. The induced transformations form the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_p)$, which preserves Pauli commutation. This first stage identifies Clifford operators that differ by a Pauli; the next stage restores that information. This group is defined using linear algebra, independently of circuits and their rewrite rules. Completeness at this stage therefore requires a proof that equal transformations give derivable equations in the symplectic theory.

D.2 Normalize the Factors, Then Compose

The first step is a normal-form proof for this symplectic group. Figure 12 shows the recursive shape: a canonical block accounts for one wire, and normalization continues on the remaining wires. The formalization proves that every circuit rewrites to this shape, that each rewrite preserves its meaning, and that the normal form is uniquely determined by that meaning. Consequently, circuits denoting the same transformation rewrite to the same normal form and are equivalent. This is the normalization strategy of Section 2, instantiated with the source paper's circuit blocks; the datatypes and normalization table are in Appendix D.5.

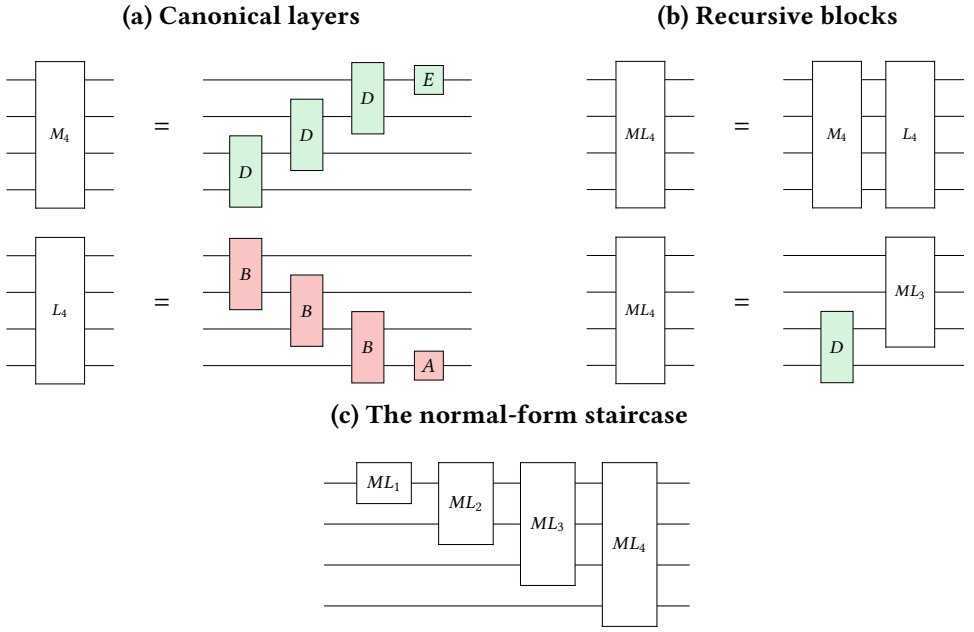


Fig. 12. The symplectic normal form, illustrated on four wires. (a) The M and L layers use the source paper's parameterized circuit blocks A, B, D, and E. (b) An ML block combines the two layers or extends a smaller ML block by a D block. (c) One block per width gives the staircase. Its uniqueness is the key completeness obligation; Paulis and scalar phases are restored separately.

Symplectic transformations retain only the Clifford action modulo Paulis. To restore the Paulis, we present their phase space $V = \mathbb{Z}_p^{2n}$ separately and combine the two presentations into the independently defined model $V \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$. The semidirect product records how a symplectic transformation acts on Pauli exponents. QCert's general composition theorem asks us to prove that

this action respects the relations of both factors. Once these obligations are checked, completeness of the combined presentation follows from completeness of the factors.

This replaces the most difficult part of following the source proof directly. Its passage from symplectic to projective completeness tracks a Pauli correction through each rewrite. We instead discharge the compatibility obligations once and use the framework's semidirect-product theorem. Verified translations then recover the paper's generators and relations. Finally, the scalar-extension theorem restores global phases, with the necessary phase corrections checked explicitly (Figure 5). The construction-specific calculations remain in the case study; the composition theorems can be reused by other developments.

D.3 What the Formalization Establishes

Agda checks the presentations of the symplectic, projective, and scalar-extension models for every odd prime p and every width n, with no postulates or unfinished proofs. In particular, it checks both normalization and the translations back to the source rule sets. This demonstrates how QCert can support a proof that needs substantial mathematical restructuring while keeping each obligation explicit.

The semantic boundary is also explicit. The Agda development presents independently constructed algebraic models; identifying them with the unitary Clifford groups and matching the paper's gate phases uses external mathematics. The Lean development proves soundness and completeness directly against complex matrices (Appendix C). The rest of this appendix gives the details. We first fix the gate conventions and state the formal theorems (Appendix D.4). We then prove symplectic completeness by normalization (Appendix D.5), restore Paulis and recover the source rule sets (Appendix D.6), and restore scalar phases (Appendix D.7). Appendix D.8 records the implementation and the boundary between the Agda proof and the external identification with unitary groups. Thus the case study shows both how the proof is assembled and exactly which connection each proof assistant checks.

D.4 Gate Conventions, Syntax, and Theorem Statements

We use the gate conventions of Bian, Li et al. [12]. Fix an odd prime p and let $\omega = e^{2\pi i/p}$. On the basis $|0\rangle, \ldots, |p-1\rangle$ indexed by $\mathbb{Z}_p$, the gates $X|j\rangle = |j+1\rangle$ and $Z|j\rangle = \omega^j|j\rangle$ generate, with the phases ω^t, the Pauli group $\mathcal{P}_n$ on n qudits. The gates

$$H\colon |j\rangle \mapsto \frac{\delta_p}{\sqrt{p}} \sum_k \omega^{jk}|k\rangle, \qquad S\colon |j\rangle \mapsto \omega^{j(j-1)/2}|j\rangle, \qquad CZ\colon |j\rangle|l\rangle \mapsto \omega^{jl}|j\rangle|l\rangle$$

generate, with the same phases, the Clifford group C_n. Here $\delta_p = (-i)^{(p-1)/2}$, which makes the determinant of H equal to 1. Modulo scalars, we write $\overline{C}_n = C_n/\langle\omega\rangle$ and $\overline{\mathcal{P}}_n = \mathcal{P}_n/\langle\omega\rangle \cong \mathbb{Z}_p^{2n}$. Then $C_n/\mathcal{P}_n \cong \overline{C}_n/\overline{\mathcal{P}}_n \cong \mathrm{Sp}(2n, \mathbb{Z}_p)$ and $\overline{C}_n \cong \overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$. We do not formalize these last two isomorphisms, because our semantic groups are *defined* as their right-hand sides. Bian, Li et al. [12] also include -1 among the global phases, so that the swap and the multiplier gates belong to C_n. We work mainly in the projective quotient, where -1 makes no difference and the relations are slightly simpler.

Using the framework of Section 3, we encode circuits, rules, and reasoning with the rules. Figure 11 shows the diagrams for the following encodings. We draw circuits in matrix-multiplication order, the reverse of the usual circuit order, so $w \bullet v$ applies v first but draws it on the right. For example, this is the swap:

```
Ex : Word (Gen (₂₊ n))
Ex = CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑ • CZ • H ↓ • H ↑
```

A rule set is an inductive family indexed by the width, with one constructor per rule. Each rule is stated once, on the bottom wires of every width at which it fits. For example, these are two of the fifteen constructors of our primary rule set _SRel,_===_, the order of the swap and the rule moving H across it:

```
data _SRel,_===_ : (n : ℕ) → WRel (Gen n) where
  order-Ex     : (₂₊ n) SRel, Ex ^ 2 === ε
  semi-Ex-H↑ : (₂₊ n) SRel, Ex • H ↑ === H ↓ • Ex
```

Reasoning with the rules is equational reasoning in the congruence ≈ they generate, with combinators for symmetry, associativity, and rewriting the left or right factor of a product. For example, conjugating by the swap moves H down a wire:

```
conj-Ex-H↑ : Ex • H ↑ • Ex ≈ H ↓
conj-Ex-H↑ = begin
  Ex • H ↑ • Ex   ≈⟨ sym assoc ⟩
  (Ex • H ↑) • Ex ≈⟨ cleft axiom semi-Ex-H↑ ⟩
  (H ↓ • Ex) • Ex ≈⟨ assoc ⟩
  H ↓ • Ex • Ex   ≈⟨ cright axiom order-Ex ⟩
  H ↓ • ε         ≈⟨ right-unit ⟩
  H ↓ ■
```

Theorem D.1 (Paper-V1.presentation). *For every width n, the fifteen rules of* `Paper-V1` *(Section D.6) present the projective Clifford model:*

```
presentation : ∀ n → (PapV1.Clifford-Relations._QRel,_===_ n) IsPresentationOf (Pauli⋊Sp n)
```

These rules are the relations of Bian, Li et al. [12] read modulo scalars, with its rule (C4) in an equivalent form. The relation _QRel,_===_ closes _SRel,_===_ under the structural rules (Section 3), and Pauli⋊Sp n is the semidirect product of $\mathbb{Z}_p^{2n}$ by $\mathrm{Sp}(2n, \mathbb{Z}_p)$ (Section D.6). The record _IsPresentationOf_ (Section 3) gives a group isomorphism from circuits modulo ≈ to $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$. That this map is well defined is *soundness*, since each rule holds. That it is injective is *completeness*, and that it is surjective says that every element of the model is represented by a circuit.

We refine the normal form of Bian, Li et al. [12] for symplectic operators, which are Clifford operators modulo Paulis, into a doubly inductive definition (Section D.5), and we prove that normal forms are unique:

Theorem D.2 (unique-nf). *Normal forms with equal symplectic denotations are equal:*

```
unique-nf : ∀ n {u v : NF n} → [[ ⌈ u ⌉ ]] ≈ˢ [[ ⌈ v ⌉ ]] → u ≡ v
```

The Route. We mostly follow the paper. Its normal boxes, its symplectic normal form, and its parametrized *box relations* reappear below. A box relation says that pushing a gate through a normal box yields an updated box and residual *dirty* gates. We also keep the reduction to eighteen relations of its Theorem 4.4, a step the paper already delegated to Agda. The one major divergence is the step that promotes symplectic to projective completeness, its Proposition 4.9. There, the paper corrects each rule $C_l = C_r$ to $C_l = C_r\,;P$ for the unique Pauli P that makes it projectively sound. It then replays the symplectic rewrites, pushing each correction to the end, where it becomes "a potentially different Pauli". We did not see how to verify this argument at reasonable cost. Instead, we work in a syntax where Paulis are trivial, since symplectic circuits have only H, S, and CZ and denote $\mathrm{Sp}(2n, \mathbb{Z}_p)$ itself. We then reintroduce Paulis and scalars by construction (Figure 13). The Pauli bookkeeping of Proposition 4.9 is thus done once, by a theorem about semidirect products that the framework proves for arbitrary groups.

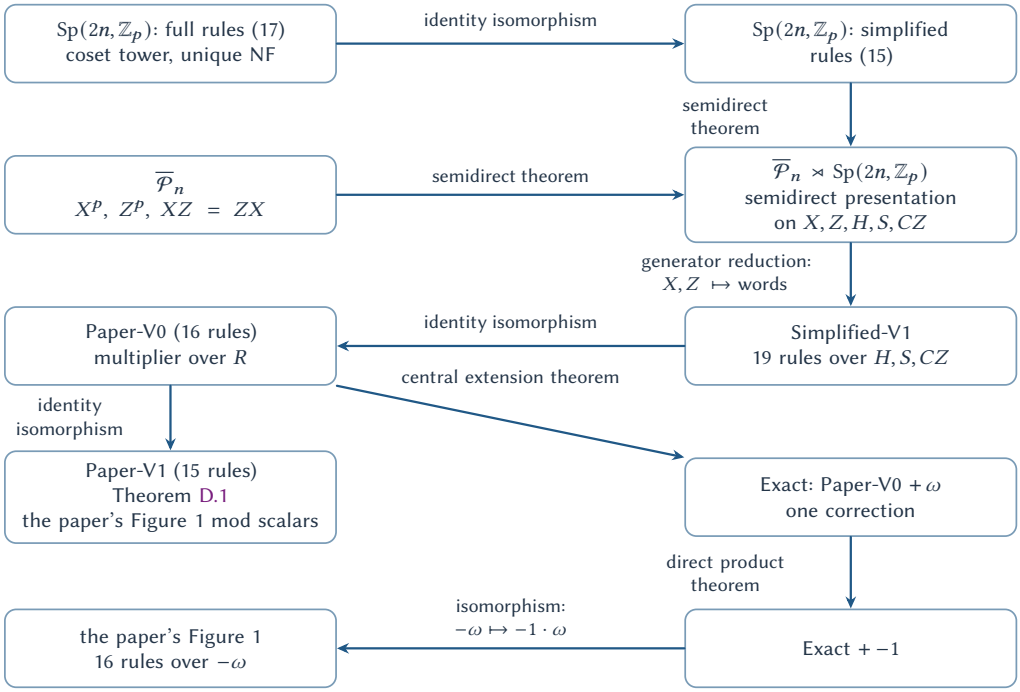


Fig. 13. The chain of presentations used to formalize the qudit Clifford result of Bian, Li et al. [12]. Every arrow is a verified presentation theorem or isomorphism of presentations. The symplectic stages present $\mathrm{Sp}(2n, \mathbb{Z}_p)$, and the projective chain from the semidirect presentation to Paper-V1 presents $\overline{\mathcal{P}}_n \rtimes \mathrm{Sp}(2n, \mathbb{Z}_p)$. "Exact" is a central extension by $\langle \omega \rangle$, and the final two stages also include the direct factor $\langle -1 \rangle$.

D.5 Normal Forms for the Symplectic Factor

We now give the datatypes and proof obligations behind Figure 12. We model $\mathrm{Sp}(2n, \mathbb{Z}_p)$ as the group of $\mathbb{Z}_p$-linear bijections of the phase space Pauli n = Vec $(\mathbb{Z}_p \times \mathbb{Z}_p)$ n that preserve the symplectic form. An element of the phase space lists, for each wire, the exponents of $X^a Z^b$ up to phase. Against this semantics, completeness rests on the normal form, which we refine into a doubly inductive datatype:

```
M L ML : ℕ → Set
M 0 = ⊤
M (1+ n) = Vec D n × E
L 0 = ⊤
L (1+ n) = Vec B n × A
ML 0 = ⊤
ML 1 = M 1 × L 1
ML m@(2+ n) = M m × L m
            ⊎ D × ML (1+ n)
```

An M-row is a column of D boxes ending in an E box, the paper's X-normal layer. An L-row is a column of B boxes ending in an A box, its Z-normal layer, with the A box at the bottom. An ML box at width $m \geq 2$ is either a full pair of rows or a bottom D box followed by an ML box one wire

up, so it is a spine of D boxes capped by a pair of rows. This is the first induction, on the width of a single coset representative. The second induction is the staircase:

```
NF : ℕ → Set
NF 0 = ⊤
NF (1+ n) = NF n × ML (1+ n)

[_] : ∀ {n} → NF n → Word (Gen n)
[_] {0}     tt       = ε
[_] {1+ n} (nf , ml) = [ nf ] ↑ • [ ml ]ᵐˡ
```

The Table. The coset table is the paper's "push a gate through a normal box", written as a structurally recursive function ract with its soundness law ract-sound (about 5,600 lines together):

```
ract       : ML (1+ n) → Gen (1+ n) → Circuit n × ML (1+ n)
ract-sound : ∀ (c : ML (1+ n)) g → let (b' , c') = ract c g in [ c ]ᶜ • [ g ]ʷ ≈ b' ↑ • [ c' ]ᶜ
```

Each clause of ract-sound is derived from the fifteen simplified symplectic relations. For instance, pushing S through an E box is one step on powers of S, namely $[\,b\,]^{\mathrm{e}} \bullet S = S^{-b} \bullet S \approx S^{-(b-1)} = [\,b-1\,]^{\mathrm{e}}$. A circuit is rewritten to its normal form by repeating this step gate by gate, so every rewrite is traced to a relation.

Uniqueness. The symplectic group acts on Vec Pauli$_1$ n, a $2n$-dimensional symplectic $\mathbb{Z}_p$-vector space in which Pauli$_1$ is the two-dimensional symplectic subspace of a single wire. The elements that fix the bottom wire's subspace pointwise form a copy of $\mathrm{Sp}(2(n-1),\mathbb{Z}_p)$ acting above the bottom wire, and the *ML* boxes represent its cosets. Every $f \in \mathrm{Sp}(2n,\mathbb{Z}_p)$ factors as $f = f' \circ c_n$, with $c_n \in ML_n$ and f' acting above the bottom wire, and c_n is unique because the image of the bottom wire determines the box. Recursing on f' writes f uniquely as $c_1 \circ c_2 \circ \cdots \circ c_n$ with $c_k \in ML_k$, which is Theorem D.2.

Surjectivity. Every symplectic transformation is the action of some circuit. We prove this constructively by clearing the bottom wire and recursing on the rest. This is the paper's Lemma 3.7, with the coset box computed rather than asserted:

```
Theorem-LM : ∀ {n} (p q : Pauli n) → sform p q ≡ ₁ → ∃ λ (lm : ML n) → act [ lm ]ᵐˡ p ≡ pZ₀ × act [ lm ]ᵐˡ q ≡ pX₀
```

Completeness. Circuits with the same denotation rewrite through the table to their normal forms. These normal forms denote the same map by soundness, so they are equal by Theorem D.2, and the circuits are equivalent. This argument is the framework's by-normalization (Figure 3a). With surjectivity, it shows that the simplified relations, which their own module also calls _SRel,_===_, present the symplectic group:

```
presentation : ∀ n → (n SRel,_===_) IsPresentationOf (Sp-group n)
```

D.6 Restoring Paulis and the Source Rule Sets

$\overline{\mathcal{P}}_n \cong (\mathbb{Z}_p \times \mathbb{Z}_p)^n$ is presented over X and Z on each wire by $X^p = Z^p = \varepsilon$ and $ZX = XZ$. Its semantic group is the phase space that $\mathrm{Sp}(2n,\mathbb{Z}_p)$ acts on, and its normal form is $X^a Z^b$ on each wire. Combining this with the symplectic presentation of Section D.5 is an instance of the framework's theorem for semidirect products. The resulting presentation has the relations of both factors together with the conjugation relations $h\,n\,h^{-1} = \mathit{conj}\,h\,n$, where the action conj spells out the paper's Lemmas 2.24–2.25. The theorem asks that the action respect the relations of each factor, and we prove this semantically. The conjugates of congruent words have equal denotations by soundness, so they are congruent by Pauli *completeness*. For symplectic rules this is respects-Δ, and

it corresponds to the paper's "Pauli complete" rules (its Definition 4.7). Since every premise is a theorem, the semidirect presentation of Pauli⋊Sp is unconditional.

Reaching the Paper's Rules. The semidirect presentation has generators X, Z, H, S, and CZ and $3 + 15 + 14$ relations (Pauli, symplectic, and conjugation), whereas the paper's rule sets have only H, S, and CZ. Three isomorphisms of presentations close this gap (Figure 13), through `Simplified-V1` (nineteen rules), `Paper-V0` (sixteen), and `Paper-V1` (the fifteen of Theorem D.1). Along the way, X and Z become words, and the Pauli and conjugation rules become theorems.

The first isomorphism, from the semidirect presentation to `Simplified-V1`, is between different alphabets. Its map f sends S to $R = S \bullet Z^{1/2}$, and its inverse h sends S to $Z^{-1/2} \bullet S$. In particular, f induces the embedding of the symplectic group into the projective Clifford group. The nineteen rules of `Simplified-V1` are the fifteen simplified symplectic rules together with $(SH)^3 = \varepsilon$, $H^2SH^2S = SH^2SH^2$, and the two rules pushing X through CZ. From these, f-well-defined derives the images under f of the three Pauli rules and the fourteen conjugation rules.

The second isomorphism, from `Simplified-V1` to `Paper-V0`, is the identity on circuits. The two rule sets share six rules word for word and three more up to the spelling of the multiplier, and each derives the rest of the other. The larger direction is `Paper-V0`'s module `Lemmas`, of about 4,750 lines.

The third isomorphism, from `Paper-V0` to `Paper-V1`, changes how the multiplier $M_x \colon |j\rangle \mapsto |xj\rangle$ is spelled as a word. `Paper-V0` spells it over R, as $M_x = R^{x^{-1}}HR^xHR^{x^{-1}}H$. The paper's (T1) spells it over S, up to a phase, as $M_x =$ SHS' $x\ x^{-1}$, where SHS' $a\ b = Z^{(b-1)/2}X^{(1-a)/2}S^bHS^aHS^bH$. Over S, the rule semi-MR reads $M_gS^{g^2}Z^{(g^2-g)/2} = SM_g$. `Paper-V1` restates the four multiplier rules over SHS' and drops one rule. Since M_1 is now the word $(SH)^3$, M-power at $k = 0$ already gives $(SH)^3 \approx \varepsilon$. Over R, M_1 is the different word $(RH)^3$, so that axiom had to stay. The bridge $R^aHR^bHR^aH \approx$ SHS' $b\ a$ for $ab = 1$ (`Shared.PauliBase`) carries the multiplier rules across, which gives Theorem D.1. The paper's own rule (C4) states the conjugation of S by M_g in the other direction. Each of (C4) and semi-MR derives the other (SemiMS), so Figure 1 of Bian, Li et al. [12] read modulo scalars is isomorphic to `Paper-V1`.

D.7 Restoring Scalar Phases

The theorem extension-presentation extends a projective presentation by a scalar. We apply it to `Paper-V0`, which has order-SH as an axiom. We add ω with $\omega^p = \varepsilon$, let ω commute with every gate, and *correct* each rule by its scalar. The correction is ε except on order-SH, the one rule that does not hold exactly, since $(SH)^3 = \omega^{-1/8}$. The exponent $-1/8$ is computed in $\mathbb{Z}_p$, where it equals $(p^2-1)/8$. The multiplier words now denote $|j\rangle \mapsto |xj\rangle$ only up to a scalar.

The semantic extension is the Weyl cocycle on the natural semidirect product, $c((P,A),(Q,B)) = \frac{1}{2}\,\mathrm{sform}(P,\ A \cdot Q)$. Here $A \cdot Q$ is the symplectic part acting on the Pauli vector, and sform is the symplectic form of the phase space, the exponent in $PQ = \omega^{\mathrm{sform}(P,Q)}\,QP$. The cocycle identity follows from the bilinearity and the symplectic invariance of sform. The cocycle of the presented group is the pullback of c along the presentation by `Paper-V0`, so the two agree by construction. The identification with unitary operators uses Weyl multiplication [37, Eq. (4), Theorem 3] and a *linear* Weil representation with exact covariance [38, Theorem 1.3.2]. The operators $U(e,v,A) = \omega^e W(v)\rho(A)$ satisfy

$$U(e,v,A)\,U(f,w,B) = U\big(e + f + \tfrac{1}{2}\mathrm{sform}(v,Aw),\ v + Aw,\ AB\big).$$

This is the Heisenberg–Weil representation [39, §1.3]. Conjugation on Weyl operators determines A and v, so the representation is faithful and its scalars are exactly $\langle \omega I\rangle$ (at width zero, $e \mapsto \omega^e$).

Adjoining $-I$ gives its direct product with $\mathbb{Z}_2$, whose scalars are $\langle -\omega I \rangle$. These finite phase choices differ from the full unitary normalizer, which contains $U(1)$. With no hypotheses, we obtain:

```
presentation-exact : ∀ n → (n Exact._===_) IsPresentationOf (PE.Presented-group n (exact-generator-data n))
```

Agda also shows that the trivial cocycle cannot realize the correction of order-SH (split-forces).

The Rules as Printed. Figure 1 of Bian, Li et al. [12] is stated over $-\omega$. Its multipliers carry Legendre symbols and its swap carries the sign $(-1)^{(p-1)/2}$, so it needs the -1 that the exact group lacks, since the scalars of that group have odd order. Adjoining -1 as a central involution presents the exact group times $\mathbb{Z}_2$, which is Clifford±. Figure 1 is isomorphic to this rule set by $-\omega \mapsto -1 \cdot \omega$, where the scalar of each exact rule is determined by reading its lift back. This is the paper's Theorem 4.10, stated against our group:

```
presentation-Figure1 : ∀ n → (n F._===_) IsPresentationOf (Clifford± n)
```

D.8 Implementation and Semantic Boundary

This development is checked by Agda 2.8 with standard library 2.4 under `--safe`, with no postulates and no holes. The development has 215 modules and about 58,000 lines of code, of which about 27,000 are for the symplectic group, 18,000 for the projective chain, and 13,000 for the scalars. These counts exclude the shared library (Appendix C).

The Agda development does not formalize the two isomorphisms of Section D.4 [37, 38], and no matrix appears in it. The representation argument above is external mathematics, and matching the paper's gate phases remains a convention check [12, 52]. The Lean development needs neither, since it proves soundness and completeness against complex matrices directly (Appendix C).

We used versions of Claude for parts of this formalization: composing box relations to prove ract-sound, defining the symplectic group, and establishing the equivalence between the code's relations and those in Figure 1 of Bian, Li et al. [12], read modulo scalars. Claude also wrote presentation-exact and the layer for Figure 1 with -1.

E Proofs for Four-Bit Reversible Circuits

This appendix proves the theorems of Section 6, following the Agda development [4]. We number the wires from the top, as in Figure 8, while the development numbers them from the bottom. A circuit acts on the bit strings $x_1x_2x_3x_4$, where x_i is the bit of wire i. We compose in execution order, for circuits and permutations alike: uw applies u and then w. The halves H_0 and H_1 are the sets of states with $x_1 = 0$ and with $x_1 = 1$. By (T3), a structural swap is three CNOTs, so we write circuits as words of gates placed on arbitrary wires. The results are labeled with their Agda identifiers.

E.1 One to Three Wires

Proposition E.1 (complete$_1$–complete$_3$, presentation$_1$–presentation$_3$). *For $k \le 3$, the theory $\mathcal{T}$ is complete for circuits on k wires, and $G_k \cong \mathrm{Sym}(2^k)$.*

Proof. On three wires, order the states along the Gray code, so that $q_0, \ldots, q_7$ are 000, 001, 011, 010, 110, 111, 101, and 100. Consecutive states differ in one bit, so their transposition $s_i = (q_i\ q_{i+1})$ is an *edge Toffoli*, a Toffoli gate onto that bit whose controls are filled or open according to the values of the other two bits. Using (T1)–(T6), every gate is an involution, and conjugating an edge Toffoli by X on a wire or by a swap of two wires gives another edge Toffoli. These symmetries of the cube carry each Coxeter relation of the s_i to one of three base cases. First, the four edge Toffolis onto one wire commute, because X on that wire is their product in two different orders, by (T2), (T4), and (T5). Second, two edge Toffolis onto different wires on disjoint edges commute, by (T6). Third,

two on edges with a common state satisfy the braid relation, by (T6) and (T3). Moreover, every gate is a product of edge Toffolis. For example, X on wire 3 is $s_4 s_2 s_6 s_0$, and the CNOT from wire 2 onto wire 3 is $s_2 s_4$. Hence every circuit is equal to the product of edge Toffolis along a Coxeter word with the same permutation. If two circuits have the same permutation, then their Coxeter words are equal by the Coxeter presentation of Sym(8) [28], and translating that derivation gives a derivation between the circuits. The cases of one and two wires are the same, with Sym(2) and Sym(4). Since every permutation is that of a Coxeter word, the semantics is onto, which gives the isomorphisms. □

On four wires, an equation whose two sides lie on the same three wires holds modulo $\mathcal{T}$ whenever it holds semantically, by Theorem E.1. We call such an equation a *three-wire step*. The development decides each one by evaluating both sides.

E.2 Reduction to Circuits That Keep a Wire

Let $N \geq 3$ and $s_d = (d \;\; d{+}1)$ for $0 \leq d \leq N-2$. The alternating group $\mathrm{Alt}(N)$ has index two in $\mathrm{Sym}(N)$, with coset representatives 1 and s_0. The Reidemeister–Schreier method [48] rewrites the Coxeter presentation of $\mathrm{Sym}(N)$ into a presentation of $\mathrm{Alt}(N)$. Its generators are $\gamma(0,d) = s_d s_0$ and $\gamma(1,d) = s_0 s_d$ for $1 \leq d \leq N-2$. Writing $\bar{c}$ for the coset other than c, its relations are

$$\gamma(c,d)\,\gamma(\bar{c},d) = 1, \qquad \gamma(\bar{c},1) = \gamma(c,1)^2,$$
$$\gamma(c,d)\,\gamma(\bar{c},d{+}1)\,\gamma(c,d) = \gamma(c,d{+}1)\,\gamma(\bar{c},d)\,\gamma(c,d{+}1), \qquad \gamma(c,e) = \gamma(\bar{c},e) \quad (e \geq 2),$$
$$\gamma(c,d)\,\gamma(\bar{c},e) = \gamma(c,e)\,\gamma(\bar{c},d) \quad (e \geq d+2),$$

for $c \in \{0,1\}$ and indices between 1 and $N-2$. They are the Coxeter relations rewritten from either coset. For $N = 16$ there are 28 generators and 238 relations, and for $N = 8$ there are 12 and 54.

Proposition E.2 (RS-complete). *Two words in the generators $\gamma(c,d)$ whose products are the same permutation are equal modulo these relations.*

Proof. The framework's Reidemeister–Schreier theorem reduces this to two obligations. Rewriting a Coxeter relation from either coset must give words that are equal modulo the relations, and rewriting the Coxeter word of a generator from the coset of 1 must give that generator back. Both are proved for every N. Completeness then transfers from the Coxeter presentation of $\mathrm{Sym}(N)$. □

Number the sixteen states along the four-bit reflected Gray code $0000, 0001, 0011, 0010, \ldots, 1000$. The states numbered 0–7 then form H_0, and each s_d flips a single wire $t(d)$, with $t(d) = 4$ for even d. A breadth-first search finds, for each generator $\gamma(c,d)$ of $\mathrm{Alt}(16)$, a circuit of at most eight gates that realizes it and targets only wires 4 and $t(d)$. Each relation involves generators with at most two values of d. So its circuits target at most three wires, and they all keep one of the wires 1–3. Let K be the set of circuits none of whose gates targets wire 1, and K_m the same for wire m. We call K *faithful* if any two circuits in K with the same permutation are equal modulo $\mathcal{T}$.

Proposition E.3 (Four.complete$_4$). *If K is faithful, then $\mathcal{T}$ is complete for four-wire circuits.*

Proof. Relabeling the wires carries faithfulness from K to every K_m. The relations therefore hold for the circuits of the generators, because the two sides of each have the same permutation and lie in some K_m. Each gate is also equal to the circuit of a word in the generators. For most gates, a Coxeter word for the gate uses only transpositions s_d that do not flip some wire $m \in \{1,2,3\}$ that the gate keeps. The gate and the circuit of the word's generators then both lie in K_m, and they have the same permutation. The four remaining gates are X on wire 1, the CNOTs from wires 3 and 4 onto wire 1, and the Toffoli from wires 3 and 4 onto wire 1. Each is equal, by a three-wire step,

to a circuit of gates translated before. Hence every circuit is equal to the circuit of a word in the generators with the same permutation. Two circuits with the same permutation thus give words with the same permutation, which are equal modulo the relations by Theorem E.2, so the circuits are equal. □

E.3 Controlled Perfect Matchings

We identify each half with the eight states of wires 2–4. Let μ_0 be the perfect matching of these states that flips wire 4, and $G = \mathrm{CX}_{1\to 4}$, which acts as the identity on H_0 and as μ_0 on H_1. For a circuit Q on wires 2–4, the conjugate $Q^{-1}GQ$ acts as the identity on H_0 and as the perfect matching $\mu = Q^{-1}\mu_0 Q$ on H_1. For each of the 105 perfect matchings μ, we fix such a circuit Q_μ and call $X_\mu = Q_\mu^{-1} G Q_\mu$ a *controlled matching*.

Lemma E.4 (CFrame.W′). *If two circuits on wires 2–4 conjugate μ_0 to the same matching, then they conjugate G to equal circuits.*

Proof. It suffices to show that a circuit R on wires 2–4 whose permutation commutes with μ_0 commutes with G. Such a permutation sends the pair of states with value v on wires 2 and 3 to the pair with some value $\sigma(v)$, flipping wire 4 or not according to v. So R has the permutation of a product of edge Toffolis onto wire 4 and a circuit on wires 2 and 3, and it equals that product by Theorem E.1. Each edge Toffoli onto wire 4 commutes with G by a four-wire equation, and the circuit on wires 2 and 3 is disjoint from G. The development computes the decomposition from the permutation and checks it by evaluation. □

Let $\bar{G} = \mathrm{X}_1 G \mathrm{X}_1$, which acts as μ_0 on H_0 and as the identity on H_1. Then $G = \mathrm{X}_4\bar{G} = \bar{G}\mathrm{X}_4$, so $X_\mu = P_\mu \bar{X}_\mu = \bar{X}_\mu P_\mu$, where $P_\mu = Q_\mu^{-1}\mathrm{X}_4 Q_\mu$ applies μ on both halves and $\bar{X}_\mu = Q_\mu^{-1}\bar{G}Q_\mu$ applies it on H_0 only.

Lemma E.5 (CRel.Calculus). *For perfect matchings μ and ν:*

(1) $X_\mu X_\mu = 1$;
(2) $X_\mu X_\nu X_\mu = X_{\mu\nu\mu}$;
(3) $X_\mu X_\nu = X_{\mu\nu}$ *if* $\mu \neq \nu$ *commute and* $\mu\nu$ *is a perfect matching.*

Proof. Item 1 holds because G is an involution. For item 2, suppose first that $\bar{X}_\mu$ and X_ν commute, as they do semantically, acting on different halves. Then $X_\mu X_\nu X_\mu = P_\mu \bar{X}_\mu X_\nu \bar{X}_\mu P_\mu = P_\mu X_\nu P_\mu$. This is the conjugate of G by $Q_\nu P_\mu$, which conjugates μ_0 to $\mu\nu\mu$, so it is $X_{\mu\nu\mu}$ by Theorem E.4.

It remains to derive the commutation. The circuit $Q_\mu \bar{X}_\mu Q_\mu^{-1}$ is $\bar{G}$, and $Q_\mu X_\nu Q_\mu^{-1}$ is $X_{\nu'}$ for $\nu' = Q_\mu \nu Q_\mu^{-1}$, by Theorem E.4. A circuit on wires 2–4 whose permutation commutes with μ_0 commutes with G, by the proof of Theorem E.4, and hence with $\bar{G}$. Such circuits carry ν' to one of five base matchings, according to the cycles of the union of ν' and μ_0, which pass through 1+1+1+1, 2+2, 2+1+1, 3+1, or 4 pairs of μ_0. For the first two base matchings, the commutation is a three-wire step. For the other three, it is a four-wire derivation of 171, 496, and 187 steps, found by a search that uses (T7), three-wire steps, and earlier four-wire equations.

For item 3, a circuit on wires 2–4 carries μ and ν to a standard pair: μ_0, and the matching that flips wires 3 and 4 together. The relation for the standard pair is a three-wire step on wires 1, 3, and 4. □

The side conditions of items 2 and 3, and the matching $\mu\nu\mu$, are computed on the matchings. The development therefore states the lemma with hypotheses that it decides by evaluation.

E.4 The Controlled Alternating Group and the Faithfulness of K

Proposition E.6 (CFaithful.cfaith). *Two words of controlled matchings with the same permutation are equal modulo $\mathcal{T}$.*

Proof. The controlled matchings act on H_1 as even permutations of the eight states, and $\mathrm{Alt}(8)$ is presented by the generators $\gamma(c,d)$ with $N = 8$, the states numbered along the Gray code. For $d \geq 2$, $\gamma(c,d)$ is a double transposition $(a\ b)(e\ f)$, the product $\big((a\ e)(b\ f)\rho\big)\big((a\ f)(b\ e)\rho\big)$ of two commuting perfect matchings, where ρ is a fixed matching of the other four states. For $d = 1$, it is a three-cycle, the product of two double transpositions that share the pair (6 7). Each generator γ is thus a word $\Phi(\gamma)$ of two or four matchings, whose controlled matchings act on H_1 as γ.

A *certificate* is a sequence of moves on words of matchings. A conjugation move rewrites $\mu\nu$ to $(\mu\nu\mu)\mu$ or to $\nu(\nu\mu\nu)$. A cancellation or insertion move deletes or inserts $\mu\mu$. A Klein move merges $\mu\nu$ into the matching $\mu\nu$, or splits it, for commuting $\mu \neq \nu$ whose product is a matching. By Theorem E.5, each move is a derivation once its side condition evaluates to true. For each of the 54 relations of $\mathrm{Alt}(8)$, a search over double transpositions finds a certificate between the images under Φ of its two sides, with 412 moves in all, and Agda checks each certificate by evaluation.

Finally, every controlled matching is equal to the image under Φ of a word in the generators. For X_{μ_0}, the word is $\gamma(0,2)\gamma(0,4)\gamma(0,6)$, by an eight-move certificate. Each other matching is reached along a breadth-first tree of depth 9, by conjugating its parent with a word $\Phi(\gamma)$. Translating each controlled matching in this way, two words with the same permutation become words in the generators with the same permutation. These are equal modulo the relations by Theorem E.2, and the certificates carry the equality back to the controlled matchings. □

Proposition E.7 (KFaithful.faithful$_3$). *The set K is faithful.*

Proof. Every gate in K is equal to a word of gates on wires 2–4 and G. The CNOTs from wire 1 onto wires 2 and 3 are conjugates of G by swaps. Let a and t be two of the wires 2–4, and b the third. The Toffoli with controls on wires 1 and a and target t is $\mathrm{CX}_{1\to b}\,\mathrm{CCX}_{b,a\to t}\,\mathrm{CX}_{1\to b}\,\mathrm{CCX}_{b,a\to t}$, by a relabeling of the four-wire equation by which a CNOT onto a control of a Toffoli conjugates it into the product of two Toffolis, and a commutation. A gate g on wires 2–4 carries each controlled matching to another, $gX_\mu = X_{g\mu g^{-1}}g$, by Theorem E.4. The development decides these 12×105 equations by evaluation. Since $G = X_{\mu_0}$, every circuit in K is equal to $X_{\mu_1}\cdots X_{\mu_r}P$ for some matchings μ_i and a circuit P on wires 2–4.

Suppose that two such circuits have the same permutation. The controlled matchings fix H_0, so the two circuits P agree on H_0. They do not involve wire 1, so they agree everywhere and are equal by Theorem E.1. Cancelling them, the two words of controlled matchings have the same permutation, so they are equal by Theorem E.6. □

Theorems E.3 and E.7 prove Theorem 6.1 on four wires. Every four-wire circuit is equal to the circuit of a word in the generators of $\mathrm{Alt}(16)$, so its permutation is even. Conversely, every even permutation is the product of a word in the generators, which the circuits of the generators realize. This gives $G_4 \cong \mathrm{Alt}(16)$ (presentation$_4$).

E.5 The Block Normal Form

The development fixes the circuits of the normal form as follows. $\mathrm{D}(B)$ is the product of edge Toffolis on wires 2–4 along a fixed Coxeter word for B, so it applies B on both halves. $\mathrm{C}(E)$ is the word of controlled matchings $\Phi(w)$ for a fixed word w in the generators of $\mathrm{Alt}(8)$ with permutation E. The circuit r_S consists of the circuits of generators of $\mathrm{Alt}(16)$ for the following permutation. It sends the states of S, in increasing order, to the states numbered 0–7, which form H_0, and the remaining

states, in increasing order, to those numbered 8–15. If that permutation is odd, it is followed by the transposition of the states numbered 14 and 15. These choices differ from those of the exposition we are preparing, but the theorems hold for any fixed choice.

Proposition E.8 (unique, onto). *The map* $(E, B, S) \mapsto [\![\mathrm{NF}(E, B, S)]\!]$ *is a bijection from* $\mathrm{Alt}(8) \times \mathrm{Sym}(8) \times \binom{\mathbb{F}_2^4}{8}$ *onto* $\mathrm{Alt}(16)$.

Proof. Let $p = [\![\mathrm{NF}(E, B, S)]\!]$. The circuits $\mathrm{D}(B)$ and $\mathrm{C}(E)$ preserve the halves, so $S = p^{-1}(H_0)$, which determines S and hence r_S. The permutation $r_S^{-1} p$ acts as B on H_0 and as BE on H_1, which determine B and E.

Conversely, let $p \in \mathrm{Alt}(16)$ and $S = p^{-1}(H_0)$, a set of eight states. The permutation $r_S^{-1} p$ preserves the halves; let it act as B_0 on H_0 and as B_1 on H_1. Let $E = B_0^{-1} B_1$. The permutation $[\![\mathrm{D}(B_0)]\!]^{-1} r_S^{-1} p$ is even, since p is and circuits are, and it acts as the identity on H_0 and as E on H_1, so E is even. Then $p = [\![\mathrm{NF}(E, B_0, S)]\!]$. □

By Theorem 6.1, every four-wire circuit u is therefore equal to the normal form of the coordinates of $[\![u]\!]$ (normal-form), and to no other normal form (unique).

Proposition E.9 (absorption, NormalForm.Absorb). *Let w be a circuit and $\mathrm{NF}(E, B, S)$ a normal form. Let $S' = w^{-1}(S)$, and let $\mathrm{diag}(A_0, A_1)$ be the permutation $r_{S'}^{-1}\, w\, r_S$, which acts as A_0 on H_0 and as A_1 on H_1. Then the circuit that applies w and then $\mathrm{NF}(E, B, S)$ is equal modulo $\mathcal{T}$ to $\mathrm{NF}(E', B', S')$, where*

$$B' = A_0 B \qquad \text{and} \qquad E' = B^{-1} A_0^{-1} A_1 B E.$$

In composition order, as in the exposition we are preparing, these read $B' = BA_0$ and $E' = EBA_1A_0^{-1}B^{-1}$.

Proof. The permutation $r_{S'}^{-1}\, w\, r_S$ preserves the halves, because $r_{S'}$ sends $S' = w^{-1}(S)$ onto H_0, w sends S' onto S, and r_S sends S onto H_0. So the composite has the permutation

$$[\![w\, \mathrm{NF}(E, B, S)]\!] = r_{S'}\, \mathrm{diag}(A_0, A_1)\, [\![\mathrm{D}(B)\, \mathrm{C}(E)]\!].$$

Its coordinates are therefore S', the block A_0B on H_0, and the block A_1BE on H_1, which give B' and E'. The derivation then comes from Theorem 6.1. The development states the formulas pointwise, as $B'(x) = B(A_0(x))$ and $E'(B'(x)) = E(B(A_1(x)))$ for the states x of wires 2–4, in function notation. □

E.6 Implementation

The development consists of 40 modules in `Examples/Groups/XCXCCX-4bit`, all reached from MainTheorem. Given the interfaces of the shared library, Agda checks them in 85 seconds, the sum of the times it reports for the modules (98 seconds of wall time, with a peak resident size of 1.9 GB). Scripts in the development generate the long derivations, namely the four-wire equations and the three base commutations of Theorem E.5, together with the tables of matchings and the 412 moves of the certificates. Agda checks each derivation step and each move by evaluation. Two traps took most of the debugging time, and both made Agda evaluate the symbolic word that the library's symmetric group development builds for an arbitrary permutation. The first is comparing two spellings of the sign of a permutation at a numeric width, such as 16 against 8 + 8. The second is meeting a goal about a circuit with a lemma about its unfolding. The development therefore reasons about signs at variable widths and keeps the normal form's building blocks opaque, with their definitions given as equations. Its Lean translation, `QCert/XCXCCX4bit`, makes the same choices, and Lean checks the same derivations and certificates in the kernel (Appendix C).